

URiCA 2026

Upcoming Researchers in
Commutative Algebra

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Conference Notes

URiCA 2026

Continuing the mission of KUMUNUjr

Inspired by the success of the KUMUNU commutative algebra conferences (KU, MU, NU), KUMUNUjr brought together graduate students and postdocs studying commutative algebra in the midwest to share their research and expertise while building lasting collaborations and friendships.

Upcoming Researchers in Commutative Algebra (URiCA) is the next chapter for KUMUNUjr.

Contents

1	Karl Schwede — New Perspectives on the Briançon–Skoda Theorem	1
1.1	Integral closure: the opening example	1
1.2	Classical Briançon–Skoda containments	2
1.3	Pseudo-rational singularities and the blowup viewpoint	4
1.3.1	Normalized blowup and integral closure	4
1.4	Derived global sections and the L -complex	5
1.4.1	The derived vanishing on the blowup	6
1.5	Pseudo-rationality as a splitting mechanism	7
1.6	Alterations and the uniform Briançon–Skoda direction	9
	Summary of the Lecture	10
2	Suchitra Pande — The Geometry of the F-Signature	12
2.1	Frobenius and the F -signature	12
2.2	Fano varieties and anticanonical section rings	14
2.3	The F -normalized volume	15
2.4	Applications	16
2.5	Proof ideas: Artinian Gorenstein algebras	16
3	Dorian Kalir — Systems of Higher Homotopies and Koszul Homomorphisms	19
3.1	Systems of higher homotopies as twisting morphisms	19
3.2	The classical complete-intersection construction	20
3.3	Resolution transfer	21
3.4	Koszul duality and the Priddy coalgebra	21
3.5	Koszul homomorphisms and universal resolution transfer	23
3.6	Exact zero-divisors	24
	Summary of the Lecture	25
4	Dalena Vien — Edge Ideals via Independence Polynomials of Graphs	26
4.1	Introduction and Motivation	26
4.2	Edge Ideals and Algebraic Invariants	26
4.2.1	Edge ideals	27
4.2.2	Minimal graded free resolutions	27
4.3	Independence Complexes and Hochster’s Formula	28
4.3.1	The independence complex	28
4.3.2	Stanley–Reisner translation	29
4.3.3	Fröberg’s theorem as a model application	29

4.4	Hilbert Series and the h -Polynomial	30
4.4.1	Hilbert series from the resolution	30
4.4.2	The f -vector and h -vector	30
4.4.3	What was known about $\deg h$	31
4.5	From Independent Sets to the Degree of the h -Polynomial	31
4.5.1	Expressing the h -polynomial using $\Delta(G)$	31
4.5.2	The top coefficient and the reduced Euler characteristic	32
4.5.3	The independence polynomial	32
4.5.4	Multiplicity of the root -1	32
4.5.5	Proof of the degree formula	33
4.5.6	Examples	34
4.6	Pseudo-Gorenstein and Pseudo-Gorenstein*	35
4.6.1	Related notions	35
4.6.2	Paths and cycles	36
4.7	Graph Suspensions	36
4.7.1	C -suspensions	36
4.7.2	Suspending over a vertex cover	37
4.7.3	Extremal values of s	38
4.7.4	Full suspensions	39
4.8	Key Takeaways and Future Directions	39
	Selected Attributions	40
5	Sean Grate – The graded Möbius algebra of a matroid	41
5.1	Free resolutions and numerical invariants	41
5.2	Koszul algebras	42
5.2.1	When does quadratic imply Koszul?	43
5.2.2	Restrictions imposed by Koszulness	44
5.3	Matroids and geometric lattices	44
5.3.1	A running example: the Fano matroid	45
5.3.2	Geometric lattices	46
5.3.3	Comparing the ranks	47
5.4	The graded Möbius algebra	48
5.4.1	A presentation by circuits	49
5.4.2	The graphic case and the Orlik–Solomon parallel	49
5.5	Modularity, Gorensteinness, and projective geometries	50
5.6	Affine geometries	51
5.7	Further directions	52
6	Jelena Mojsilović — Residual Intersections and Their Connections to Linkage	53
6.1	Background and motivation	53
6.2	Basic definitions	53

6.3	Linkage and the Peskine–Szpiro theorem	53
6.4	The Artin–Nagata condition and residual S_2 properties	54
6.5	Determinantal examples	54
6.6	Generators of residual intersections	55
6.7	$(g + 1)$ -residual intersections as sums of links	56
6.8	Closing perspective	57
7	Zach Nason — Quasi-Gorenstein Morphisms over DG-Rings	58
7.1	Classical quasi-Gorenstein homomorphisms	58
7.2	DG-rings and DG-modules	59
7.3	The derived category and derived functors	60
7.4	Projective dimension and Gorenstein dimension	60
7.5	Quasi-Gorenstein morphisms of DG-rings	61
7.6	Thick subcategories and virtual smallness	62
7.7	The complete-intersection analogy	63
7.8	The main characterization	63
	Summary of the Lecture	64
8	Maria Akter — Characterizations of Minimal Frobenius Splitting Numbers in Gorenstein Rings	65
8.1	Frobenius and restriction of scalars	65
8.2	F -purity, strong F -regularity, and splitting numbers	66
8.3	Splitting ideals and the splitting prime	67
8.4	The Gorenstein colon formula	68
8.5	Hypersurface quotients	69
8.6	Minimal Frobenius splitting numbers	70
8.7	A hypersurface criterion	72
8.8	Perturbations and continuity	73
8.9	Why the Gorenstein hypothesis matters	74
	Summary of the Lecture	75

1 Karl Schwede — New Perspectives on the Briançon–Skoda Theorem

Karl Schwede

Integral closure, birational geometry, and uniform Briançon–Skoda phenomena

Abstract

The Briançon–Skoda theorem compares integral closures of high powers of an ideal with ordinary powers. The discussion begins with the basic example $(x^2, y^2) \subseteq \mathbb{C}\{x, y\}$, passes to normalized blowups and derived global sections, and explains how splitting phenomena associated with pseudo-rational singularities lead to Briançon–Skoda containments. The final part places the argument in a uniform birational framework using alterations and a multiplier that is independent of the ideal.

1.1 Integral closure: the opening example

The lecture begins with the simplest example illustrating why ordinary ideal membership can be too rigid.

Example 1.1. Consider the convergent power-series ring

$$R = \mathbb{C}\{x, y\}$$

and the ideal

$$I = (x^2, y^2) \subseteq R.$$

Then

$$xy \notin I,$$

but, in the language of the lecture, it “feels” as though xy should belong to a natural enlargement of I .

There are two reasons for this.

(1) **Analytic evidence.** For every $(x, y) \in \mathbb{C}^2$,

$$|xy| \leq \max\{|x|^2, |y|^2\}.$$

Indeed, if $|x| \geq |y|$, then $|xy| \leq |x|^2$, while if $|y| \geq |x|$, then $|xy| \leq |y|^2$.

(2) **Algebraic evidence.** We have

$$(xy)^2 = x^2y^2 \in I^2,$$

since

$$I^2 = (x^2, y^2)(x^2, y^2) = (x^4, x^2y^2, y^4).$$

The appropriate enlargement is the *integral closure* of an ideal.

Definition 1.2 (Integral closure in the analytic local setting). Let

$$J = (f_1, \dots, f_n) \subseteq \mathbb{C}\{x_1, \dots, x_d\}$$

and let $h \in \mathbb{C}\{x_1, \dots, x_d\}$. The lecture writes $h \in \bar{J}$ if either—equivalently, both—of the following conditions holds.

(1) **Analytic condition.** There is a constant $C > 0$ such that

$$|h(z)| \leq C \max\{|f_1(z)|, \dots, |f_n(z)|\}$$

for all z in some neighborhood of the origin.

(2) **Algebraic condition.** There exists an integer $r > 0$ and elements

$$a_i \in J^i \quad (1 \leq i \leq r)$$

such that

$$h^r + a_1 h^{r-1} + a_2 h^{r-2} + \dots + a_{r-1} h + a_r = 0. \quad (1.1)$$

Remark 1.3 (How Example 1.1 fits the definition). For $I = (x^2, y^2)$ and $h = xy$, the analytic condition holds with $C = 1$. Algebraically,

$$h^2 - x^2 y^2 = 0,$$

and the constant coefficient $-x^2 y^2$ lies in I^2 . Hence

$$xy \in \bar{I} \setminus I.$$

This is the first instance of the general principle that integral closure detects elements that have the same asymptotic size, or the same birational order of vanishing, as the ideal.

1.2 Classical Briançon–Skoda containments

The lecture next records the classical comparison between integral closures of high powers and ordinary powers.

Theorem 1.4 (Briançon–Skoda). *Let*

$$J = (f_1, \dots, f_n) \subseteq \mathbb{C}\{x_1, \dots, x_d\}.$$

Then for every integer $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.2)$$

There is also the dimension form

$$\overline{J^{d+k-1}} \subseteq J^k \quad (k \geq 1). \quad (1.3)$$

In particular, taking $k = 1$ in (1.2) gives

$$\overline{J^n} \subseteq J.$$

Remark 1.5 (What the exponent shift means). The point is not merely that a sufficiently high power of J is contained in J^k ; that is automatic. The theorem says that even after enlarging J^{n+k-1} to its integral closure, no element escapes J^k . Thus integral dependence can lower the effective exponent by at most $n - 1$ in the n -generated form. When both (1.2) and (1.3) are available, one may summarize them as the stronger bound obtained from the smaller of n and d .

The analytic theorem was then placed in a purely algebraic regular-ring setting.

Theorem 1.6 (Lipman–Sathaye, 1981). *Let R be a regular ring and let*

$$J = (f_1, \dots, f_n) \subseteq R.$$

Then for every $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.4)$$

One also has the dimension bound

$$\overline{J^{\dim R + k - 1}} \subseteq J^k \quad (k \geq 1). \quad (1.5)$$

The next step is to permit singularities, at the cost of measuring the exponent by the dimension.

Theorem 1.7 (Lipman–Teissier, 1981). *Let $(R, \mathfrak{m})$ be a local pseudo-rational ring of dimension d , and let $J \subseteq R$ be any ideal. Then for every $k \geq 1$,*

$$\overline{J^{d+k-1}} \subseteq J^k. \quad (1.6)$$

Remark 1.8 (Regular versus pseudo-rational). Regular rings admit the sharper bound in terms of the number of generators of J . Pseudo-rational rings may be singular, and the classical Lipman–Teissier statement uses $d = \dim R$. The derived argument below recovers the full n -generated exponent under an appropriate splitting property associated with pseudo-rationality.

1.3 Pseudo-rational singularities and the blowup viewpoint

The lecture lists the following examples.

Example 1.9 (Examples recorded in the lecture). (1) Regular rings are pseudo-rational.

(2) The quadric cone

$$\mathbb{k}[x, y, z]/(x^2 + y^2 - z^2)$$

is pseudo-rational when $\text{char}(\mathbb{k}) \neq 2$.

(3) The cubic cone

$$\mathbb{k}[x, y, z]/(x^3 + y^3 - z^3)$$

is recorded as never pseudo-rational.

The next change of viewpoint is geometric: think of elements of R as functions on $\text{Spec } R$ and modify $\text{Spec } R$ so that the ideal becomes locally principal.

1.3.1 Normalized blowup and integral closure

Let $J \subseteq R$ be an ideal, and let

$$\pi: Y \longrightarrow \text{Spec } R$$

be the normalized blowup of J in the setting where this normalization is defined. On the standard affine charts of Y , the pulled-back ideal $J\mathcal{O}_Y$ is principal (equivalently, invertible).

The opening example reappears very concretely.

Example 1.10 (The normalized blowup of (x^2, y^2)). Take

$$I = (x^2, y^2) \subseteq \mathbb{C}[x, y].$$

The normalized blowup is covered by the familiar charts

$$\mathbb{C}\left[\frac{x}{y}, y\right] \quad \text{and} \quad \mathbb{C}\left[x, \frac{y}{x}\right].$$

On the first chart,

$$\begin{aligned} I \mathbb{C} \left[\frac{x}{y}, y \right] &= \left(\left(\frac{x}{y} \right)^2 y^2, y^2 \right) \\ &= (y^2), \end{aligned}$$

and

$$xy = \frac{x}{y} y^2 \in (y^2).$$

On the second chart,

$$\begin{aligned} I \mathbb{C} \left[x, \frac{y}{x} \right] &= \left(x^2, \left(\frac{y}{x} \right)^2 x^2 \right) \\ &= (x^2), \end{aligned}$$

and

$$xy = \frac{y}{x} x^2 \in (x^2).$$

Thus xy belongs to the pullback of I on every chart of the normalized blowup.

The lecture summarizes the normalized-blowup criterion in the formula

$$\bar{J} = \Gamma(Y, J\mathcal{O}_Y) \cap R. \quad (1.7)$$

Remark 1.11 (Interpretation of (1.7)). The formula says that an element $h \in R$ lies in $\bar{J}$ precisely when, after passing to the normalized blowup where J becomes invertible, the pullback of h is a local section of $J\mathcal{O}_Y$. In Example 1.10, this is exactly what happens to xy .

Strictly speaking, the intersection in (1.7) is taken after viewing both sides inside the appropriate total quotient ring (or fraction field in the domain case). This is the natural ambient interpretation of the intersection in (1.7).

1.4 Derived global sections and the L -complex

The lecture now replaces ordinary global sections by derived global sections. Form

$$\mathbf{R}\Gamma(Y, \mathcal{O}_Y),$$

a complex of R -modules whose cohomology is the sheaf cohomology:

$$H^i(\mathbf{R}\Gamma(Y, \mathcal{O}_Y)) \cong H^i(Y, \mathcal{O}_Y). \quad (1.8)$$

Let

$$J = (f_1, \dots, f_n), \quad \underline{f} = (f_1, \dots, f_n).$$

Instead of working only with the quotient R/J , the lecture replaces it by the Koszul complex:

$$H_0(\text{Kos}(\underline{f})) \cong R/J.$$

For higher powers, the analogous replacement is a complex denoted

$$L^k(\underline{f}),$$

with

$$H_0(L^k(\underline{f})) \cong R/J^k. \quad (1.9)$$

Remark 1.12 (Background on $L^k(\underline{f})$). The complex $L^k(\underline{f})$ is the Buchsbaum–Eisenbud L -complex associated to the k th power of the ideal generated by $\underline{f}$. For $k = 1$ it is the Koszul complex. If $f_1, \dots, f_n$ form a regular sequence, $L^k(\underline{f})$ is a free resolution of R/J^k ; in general, the feature used in the lecture is the zeroth-homology identity (1.9).

1.4.1 The derived vanishing on the blowup

Write the pulled-back ideal as

$$J\mathcal{O}_Y = \mathcal{O}_Y(-E),$$

where E denotes the effective exceptional divisor corresponding to J on the blowup.

Lemma 1.13 (The key derived vanishing). *For every $k \geq 1$, the canonical composite*

$$\mathcal{O}_Y(-(n+k-1)E) \longrightarrow \mathcal{O}_Y \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathcal{O}_Y \quad (1.10)$$

is zero in the derived category $D(Y)$.

Remark 1.14 (Why this is the decisive step). The sheaf $\mathcal{O}_Y(-(n+k-1)E)$ is the $(n+k-1)$ st power of the invertible ideal $J\mathcal{O}_Y$. Thus Lemma 1.13 says that, on the space where J has been principalized, sufficiently high order along E kills the map into the complex that detects R/J^k .

Applying derived global sections to (1.10) gives the global form of the same vanishing.

Theorem 1.15 (Global derived Briançon–Skoda mechanism). *The canonical map*

$$\mathbf{R}\Gamma(Y, \mathcal{O}_Y(-(n+k-1)E)) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \quad (1.11)$$

is zero in $D(R)$. Consequently, elements of $\overline{J^{n+k-1}}$ map to zero in

$$H_0\left(L^k(\underline{f}) \otimes_R^L \mathbf{R}\Gamma(Y, \mathcal{O}_Y)\right).$$

Proof. The first assertion is obtained by applying the derived functor $\mathbf{R}\Gamma(Y, -)$ to Lemma 1.13 and using the projection formula. For the second assertion, combine the normalized-blowup description of integral closure with the identification

$$J\mathcal{O}_Y = \mathcal{O}_Y(-E).$$

Thus an element of $\overline{J^{n+k-1}}$ gives a global section of the source of (1.11), hence has zero image in zeroth homology of the target. $\square$

Remark 1.16 (What has and has not yet been proved). At this point, the argument has not yet shown that an element of $\overline{J^{n+k-1}}$ lies in J^k . It has shown that the element dies after mapping into a derived object built from the blowup. The remaining issue is to compare that derived object back to R/J^k . This is exactly where a splitting of

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$$

enters.

1.5 Pseudo-rationality as a splitting mechanism

Pseudo-rationality enters the argument through the following derived splitting condition.

Definition 1.17 (Splitting formulation used in the lecture). In the setting under discussion, the lecture writes that R is pseudo-rational precisely when R is Cohen–Macaulay and, for every relevant blowup

$$\pi: Y \longrightarrow \operatorname{Spec} R,$$

the natural map

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \tag{1.12}$$

splits in the derived category. Thus there is a morphism

$$\varphi: \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow R$$

such that the composite

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \xrightarrow{\varphi} R$$

is the identity.

Remark 1.18 (Birational splitting). The splitting condition is closely related to the birational derived-splinter property. Under the standard hypotheses used in this setting, pseudo-rational local rings satisfy the required birational splitting, and that is the feature needed for the Briançon–Skoda argument below.

The splitting turns the derived vanishing into actual ideal membership.

Theorem 1.19 (Full n -generated Briançon–Skoda from splitting). *Let $J = (f_1, \dots, f_n) \subseteq R$, and let $Y \rightarrow \operatorname{Spec} R$ be the normalized blowup appearing above. Assume that*

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$$

admits a retraction in $\mathbf{D}(R)$. Then, for every $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.13)$$

In particular, the lecture applies this to the pseudo-rational situation.

Proof. Let

$$\varphi: \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow R$$

be a retraction. Tensoring with the perfect complex $L^k(\underline{f})$ gives a map

$$L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} R.$$

Taking zeroth homology and using (1.9) yields a comparison with R/J^k . Schematically,

$$\begin{array}{ccc} R & \xrightarrow{\quad} & H_0\left(L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y)\right) \\ & \searrow \text{mod } J^k & \downarrow H_0(1 \otimes \varphi) \\ & & H_0\left(L^k(\underline{f}) \otimes_R^{\mathbf{L}} R\right) \cong R/J^k. \end{array}$$

By Theorem 1.15, every element of $\overline{J^{n+k-1}}$ maps to zero along the top arrow. Therefore its image in R/J^k is also zero, which is exactly (1.13). $\square$

Remark 1.20 (Why this improves the dimension bound). For an n -generated ideal in a d -dimensional pseudo-rational ring, the classical statement in Theorem 1.7 uses the exponent $d + k - 1$. The derived splitting argument recovers the generator-sensitive exponent $n + k - 1$. When $n < d$, this is a genuinely stronger containment.

1.6 Alterations and the uniform Briançon–Skoda direction

The final step asks how one can retain a useful remnant of the splitting argument even when R itself need not satisfy the pseudo-rational splitting property.

Suppose

$$\eta: R \longrightarrow \mathbf{R}\Gamma(Z, \mathcal{O}_Z)$$

comes from a map

$$Z \longrightarrow \operatorname{Spec} R$$

which may be one of the following objects listed in the lecture:

- a resolution of singularities;
- a regular alteration;
- a truncated regular alteration hypercover.

The crucial geometric requirement is that this construction dominates, or factors through, the blowups used above. In derived global sections one has a factorization of the form

$$\begin{array}{ccc} & \mathbf{R}\Gamma(Y, \mathcal{O}_Y) & \\ \nearrow & & \searrow \\ R & \xrightarrow[\eta]{} & \mathbf{R}\Gamma(Z, \mathcal{O}_Z), \end{array}$$

for the relevant blowups $Y \rightarrow \operatorname{Spec} R$.

Assume in addition that there is a map

$$\phi: \mathbf{R}\Gamma(Z, \mathcal{O}_Z) \longrightarrow R \tag{1.14}$$

such that

$$\phi \circ \eta = c \cdot \operatorname{id}_R \tag{1.15}$$

for some nonzero element $c \in R$.

Remark 1.21 (Why multiplication by c replaces an honest splitting). For a pseudo-rational ring one has $c = 1$: the map back is an actual retraction. In the alteration setting one may only be able to split after passing to the generic point. Clearing denominators then produces a nonzero c for which the composite is multiplication by c . Thus c measures the failure of an honest splitting over all of R .

Combining this “splitting up to c ” with the derived vanishing gives the following containment.

Proposition 1.22 (A uniform multiplier). *Let $J = (f_1, \dots, f_n) \subseteq R$. Under the factorization and map (1.14)–(1.15) described above, the lecture obtains*

$$c \overline{J^{n+k-1}} \subseteq J^k \quad (k \geq 1). \quad (1.16)$$

The important point is that c is chosen from the ambient birational/alteration construction rather than separately from each ideal J .

Proof mechanism. Take $h \in \overline{J^{n+k-1}}$. By Theorem 1.15, its image in

$$H_0\left(L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y)\right)$$

is zero. Pass through the factorization to $\mathbf{R}\Gamma(Z, \mathcal{O}_Z)$, then apply $1 \otimes \phi$. Because the composite $R \rightarrow \mathbf{R}\Gamma(Z, \mathcal{O}_Z) \rightarrow R$ is multiplication by c , the image of h in R/J^k becomes the class of ch . Hence $ch = 0$ in R/J^k , or equivalently

$$ch \in J^k.$$

This proves (1.16). □

Remark 1.23 (From a multiplier to a uniform exponent). Passing from the existence of a uniform multiplier c in (1.16) to a uniform exponent is not formal; it uses the uniform Briançon–Skoda and Artin–Rees machinery that controls integral closures simultaneously for all ideals.

Corollary 1.24 (Uniform Briançon–Skoda). *Let R be a quasi-excellent reduced Noetherian ring of finite dimension. Then there exists an integer $\ell > 0$, depending only on R , such that for every ideal $J \subseteq R$ and every $k \geq 1$,*

$$\overline{J^{k+\ell}} \subseteq J^k. \quad (1.17)$$

Remark 1.25 (Conceptual role of uniformity). The earlier theorems bound the shift by data attached to the particular ideal—the number of generators or the dimension. The uniform statement asserts that one exponent loss ℓ , depending only on the ambient ring, works simultaneously for every ideal. This is the bridge from the Briançon–Skoda problem to uniform Artin–Rees phenomena.

Summary of the Lecture

The lecture develops the following chain of ideas.

- (i) The element xy is not in (x^2, y^2) , but both analytic size estimates and an integral equation show that it belongs to the integral closure. This motivates the analytic and algebraic definitions of $\overline{J}$.
- (ii) The Briançon–Skoda theorem compares integral closures of high powers with ordinary powers. In a regular setting, an n -generated ideal satisfies

$$\overline{J^{n+k-1}} \subseteq J^k.$$

Lipman–Teissier obtain a dimension-based version for pseudo-rational singularities.

- (iii) The normalized blowup converts J into an invertible ideal $J\mathcal{O}_Y = \mathcal{O}_Y(-E)$ and gives a geometric description of integral closure by global sections. The example (x^2, y^2) becomes principal on each normalized blowup chart, and xy lies in the pulled-back ideal on every chart.
- (iv) Replacing R/J^k by the Buchsbaum–Eisenbud complex $L^k(\underline{f})$ exposes a derived vanishing:

$$\mathbf{R}\Gamma(Y, \mathcal{O}_Y(-(n+k-1)E)) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$$

is zero. A derived splitting $R \rightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$ then turns that vanishing into the actual containment $\overline{J^{n+k-1}} \subseteq J^k$.

- (v) Finally, a resolution, regular alteration, or suitable alteration hypercover can provide a weaker retraction whose composite is multiplication by a nonzero c . This yields

$$c \overline{J^{n+k-1}} \subseteq J^k$$

uniformly in J , which is the key input behind the uniform Briançon–Skoda direction recorded at the end of the lecture.

2 Suchitra Pande — The Geometry of the F -Signature

Suchitra Pande

Joint work in progress with Yuchen Liu

Abstract

The F -signature measures the asymptotic number of free summands appearing in iterates of Frobenius. After reviewing its basic algebraic properties, the discussion turns to geometric information encoded by this invariant for anticanonical section rings of Fano varieties. A sharp inequality compares the F -signature with anticanonical volume, and equality is linked to K -semistability. The Fermat cubic surface and full flag varieties illustrate the connection. The proof strategy is driven by Artinian Gorenstein algebras and the structure of Frobenius splitting ideals.

2.1 Frobenius and the F -signature

Let k be a perfect field of characteristic $p > 0$, and let $(R, \mathfrak{m}, k)$ be an F -finite local ring essentially of finite type over k . Write

$$d = \dim R.$$

For every integer $e \geq 1$, the e th Frobenius endomorphism is

$$F^e: R \longrightarrow R, \quad r \longmapsto r^{p^e}.$$

The notation $F_*^e R$ denotes the additive group of R endowed with the R -module structure

$$r \cdot F_*^e x = F_*^e(r^{p^e} x).$$

Thus the iterates of Frobenius produce a sequence of finite R -modules

$$R, F_* R, F_*^2 R, \dots$$

Theorem 2.1 (Kunz). *The ring R is regular if and only if $F_*^e R$ is free for some, equivalently for every, $e \geq 1$.*

This motivates measuring how much of $F_*^e R$ is free even when R is singular.

Definition 2.2 (Frobenius splitting numbers and F -signature). For $e \geq 1$, write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e,$$

where M_e has no free direct summand and $a_e(R)$ is maximal. The integer $a_e(R)$ is the e th *Frobenius splitting number*. Since the residue field is perfect, the F -signature is

$$s(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{ed}}.$$

The existence of this limit is due to Tucker.

Theorem 2.3 (Basic properties of the F -signature). *For an F -finite local ring as above,*

$$0 \leq s(R) \leq 1.$$

Moreover:

- (i) $s(R) = 1$ if and only if R is regular;
- (ii) $s(R) > 0$ if and only if R is strongly F -regular.

Consequently, positivity of the F -signature forces R to be normal and Cohen–Macaulay; in the standard excellent setting it also implies pseudo-rationality.

Example 2.4 (Veronese subrings). Let

$$S = k[x_1, \dots, x_n]$$

and let $A = S^{(r)} \subseteq S$ be the r th Veronese subring. If $\mathfrak{m} = A_+$ is its homogeneous maximal ideal, then in the tame case $p \nmid r$ the corresponding quotient-singularity computation gives

$$s(A_{\mathfrak{m}}) = \frac{1}{r}.$$

Hence the F -signature can be positive while becoming arbitrarily small as $r \rightarrow \infty$.

Example 2.5 (The Fermat cubic cone). For a prime p , set

$$A_p = \frac{\mathbb{F}_p[x, y, z, w]}{(x^3 + y^3 + z^3 + w^3)}, \quad \mathfrak{m} = (x, y, z, w)A_p, \quad R_p = (A_p)_{\mathfrak{m}}.$$

For $p = 2, 3$ one has $s(R_p) = 0$. For $p \geq 5$, write $p = 3k + 1$ or $p = 3k + 2$. Then

$$s(R_p) = \frac{1}{8} - \frac{1}{4(27k^2 + 27k + 8)}.$$

In particular,

$$\lim_{p \rightarrow \infty} s(R_p) = \frac{1}{8}.$$

The projective variety

$$X_p = \text{Proj}(A_p)$$

is a smooth cubic surface for $p \neq 3$. This example already indicates that the asymptotic behavior of the local invariant $s(R_p)$ can reflect global projective geometry.

2.2 Fano varieties and anticanonical section rings

Definition 2.6 (Fano variety). Let X be a smooth projective variety of dimension n over a field. It is *Fano* if the anticanonical line bundle

$$\omega_X^{-1} \cong \bigwedge^n T_X$$

is ample.

Example 2.7 (Hypersurfaces). Let

$$X = V(f) \subseteq \mathbb{P}^{n+1}$$

be a smooth hypersurface defined by a homogeneous polynomial of degree d . Adjunction gives

$$K_X \cong \mathcal{O}_X(d - n - 2).$$

Thus:

- X is Fano when $d \leq n + 1$;
- $K_X \cong \mathcal{O}_X$ when $d = n + 2$;
- K_X is ample when $d \geq n + 3$.

For a Fano variety X , consider its anticanonical section ring

$$A(X, -K_X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(-mK_X)).$$

When discussing its F -signature, we localize at the homogeneous maximal ideal

$$\mathfrak{m} = A(X, -K_X)_+$$

and write

$$R = A(X, -K_X)_{\mathfrak{m}}.$$

We also set

$$\mathrm{vol}(X) = (-K_X)^n.$$

Theorem 2.8 (Volume bound for the F -signature). *Let X be an n -dimensional Fano variety in positive characteristic, let $A = A(X, -K_X)$, and put $R = A_{A_+}$. Then*

$$\mathrm{vol}(X) \geq 2^n(n+1)! s(R).$$

Equivalently,

$$s(R) \leq \frac{\mathrm{vol}(X)}{2^n(n+1)!}.$$

The equality case is the geometric focus.

Theorem 2.9 (Liu–Pande, work in progress). *Let X be an n -dimensional Fano variety in positive characteristic, let $A = A(X, -K_X)$, and put $R = A_{A_+}$. If*

$$s(R) = \frac{\mathrm{vol}(X)}{2^n(n+1)!},$$

then X is K -semistable.

2.3 The F -normalized volume

The preceding inequality is closely related to a Frobenius-theoretic analogue of normalized volume.

Definition 2.10 (F -normalized volume). Let $(R, \mathfrak{m})$ be a d -dimensional local ring of characteristic $p > 0$. Define

$$\widehat{\mathrm{vol}}_F(R) = \inf_{\substack{\mathfrak{a} \subseteq \mathfrak{m} \\ \mathfrak{a} \text{ is } \mathfrak{m}\text{-primary}}} \mathrm{fpt}(\mathfrak{a})^d e(\mathfrak{a}),$$

where $\mathrm{fpt}(\mathfrak{a})$ is the F -pure threshold and $e(\mathfrak{a})$ is the Hilbert–Samuel multiplicity.

The F -pure threshold measures how far powers of an ideal can be inserted into Frobenius splittings. It plays the positive-characteristic role of the log canonical threshold. Consequently, the quantity

$$\mathrm{fpt}(\mathfrak{a})^d e(\mathfrak{a})$$

is the Frobenius analogue of the expressions appearing in the normalized-volume theory of

singularities. The comparison with K -stability is therefore parallel to the characteristic-zero criterion that detects K -semistability from the singularity of the anticanonical cone.

2.4 Applications

Example 2.11 (Full flag varieties). Let V be an n -dimensional vector space over k , and let

$$\mathrm{Fl}(V)$$

be the full flag variety parametrizing chains

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V, \quad \dim V_i = i.$$

The full flag variety is Fano. For its anticanonical section ring R , the F -signature attains the extremal value

$$s(R) = \frac{\mathrm{vol}(\mathrm{Fl}(V))}{2^{\dim \mathrm{Fl}(V)} (\dim \mathrm{Fl}(V) + 1)!}.$$

The equality criterion therefore recovers the K -semistability of the flag variety.

Example 2.12 (The Fermat cubic surface). Let

$$X = V(x^3 + y^3 + z^3 + w^3) \subseteq \mathbb{P}^3.$$

Then X is a del Pezzo surface of degree 3, so

$$\mathrm{vol}(X) = (-K_X)^2 = 3.$$

Hence

$$\frac{\mathrm{vol}(X)}{2^2 \cdot 3!} = \frac{3}{24} = \frac{1}{8}.$$

Together with

$$\lim_{p \rightarrow \infty} s(R_p) = \frac{1}{8},$$

this shows that the reductions of the Fermat cubic asymptotically attain the geometric upper bound for the F -signature.

2.5 Proof ideas: Artinian Gorenstein algebras

A key mechanism is to convert free summands of Frobenius into finite-length Gorenstein quotients.

Definition 2.13 (Artinian Gorenstein algebra). A finite-dimensional local k -algebra $(A, \mathfrak{n})$ is Artinian Gorenstein if

$$\dim_k \operatorname{Ann}_A(\mathfrak{n}) = 1.$$

Equivalently, its socle is one-dimensional over k .

In the standard graded case,

$$A = A_0 \oplus A_1 \oplus \cdots \oplus A_j, \quad A_0 \cong A_j \cong k,$$

Artinian Gorenstein duality gives

$$\dim_k A_i = \dim_k A_{j-i}$$

for every i . Thus the Hilbert function is symmetric.

For a local Artinian Gorenstein algebra, the associated graded ring

$$\operatorname{gr}_{\mathfrak{n}}(A) = A/\mathfrak{n} \oplus \mathfrak{n}/\mathfrak{n}^2 \oplus \mathfrak{n}^2/\mathfrak{n}^3 \oplus \cdots$$

need not itself be Gorenstein. Nevertheless, self-duality gives, for every ideal $I \subseteq A$,

$$\ell(A/I) = \ell(\operatorname{Ann}_A(I)).$$

If $\mathfrak{n}^{j+1} = 0$, then

$$\mathfrak{n}^{j+1-i} \subseteq \operatorname{Ann}_A(\mathfrak{n}^i),$$

which yields strong “front-loaded” constraints on the Hilbert function.

The Frobenius splitting numbers admit precisely such an Artinian interpretation.

Definition 2.14 (Frobenius splitting ideal). For $e \geq 1$, define

$$I_e(R) = \{x \in R \mid \phi(F_*^e x) \in \mathfrak{m} \text{ for every } \phi \in \operatorname{Hom}_R(F_*^e R, R)\}.$$

Proposition 2.15 (Gorenstein reduction). *Assume that $(R, \mathfrak{m}, k)$ is F -finite and Gorenstein, with k perfect. Then $R/I_e(R)$ is Artinian Gorenstein and*

$$\ell_R(R/I_e(R)) = a_e(R).$$

Thus estimates for the Frobenius splitting numbers can be converted into length and Hilbert-function estimates for Artinian Gorenstein quotients.

This is the bridge between local Frobenius algebra and the global volume inequalities: one

first controls finite-length Gorenstein quotients, then passes to the asymptotic limit that defines the F -signature.

3 Dorian Kalir — Systems of Higher Homotopies and Koszul Homomorphisms

Dorian Kalir

Joint work with Kory Pollicove and Ryan Watson

Abstract

Systems of higher homotopies were introduced to transfer free resolutions across quotients by regular sequences. Their classical formulas can be encoded compactly by twisting morphisms from a coalgebra into an endomorphism algebra. Koszul duality supplies the correct coalgebraic object—the Priddy coalgebra—and leads to a general resolution-transfer theorem for Koszul homomorphisms. The final example treats a quotient by an exact zero-divisor and exhibits the higher-homotopy identities explicitly.

3.1 Systems of higher homotopies as twisting morphisms

Let F be a complex of R -modules and let C be a differential graded coalgebra. The endomorphism complex

$$\mathrm{End}_R(F)$$

is a differential graded algebra under composition.

Definition 3.1 (System of higher homotopies). A *system of higher homotopies* on F over C is a twisting morphism

$$\sigma: C \longrightarrow \mathrm{End}_R(F).$$

In the curved setting, the coalgebra C is equipped with a degree -2 R -linear map

$$h: C \longrightarrow R,$$

called the *curvature*. If

$$\eta_{\mathrm{End}}: R \longrightarrow \mathrm{End}_R(F), \quad r \longmapsto r \, \mathrm{id}_F,$$

denotes the unit map, then σ has degree -1 and satisfies the curved Maurer–Cartan equation

$$d_{\mathrm{End}}\sigma + \sigma d_C + \mu_{\mathrm{End}}(\sigma \otimes \sigma)\Delta_C = \eta_{\mathrm{End}} \circ h,$$

with the standard Koszul signs.

This formulation packages the classical Shamash–Eisenbud higher homotopies used in resolution transfer.

3.2 The classical complete-intersection construction

Let

$$S = R/(f_1, \dots, f_c),$$

where $f_1, \dots, f_c$ is an R -regular sequence, and let

$$F \xrightarrow{\sim} M$$

be a free R -resolution of an S -module M . Write

$$H = (h_1, \dots, h_c) \in \mathbb{N}^c, \quad |H| = h_1 + \dots + h_c,$$

and let e_i denote the i th standard basis vector of $\mathbb{N}^c$.

Definition 3.2 (Classical higher homotopies). A classical system of higher homotopies on F consists of maps

$$\sigma^{(H)}: F \longrightarrow F$$

of homological degree $2|H| - 1$, indexed by $H \in \mathbb{N}^c$, satisfying

$$\begin{aligned} \sigma^{(0)} &= \partial^F, \\ \partial^F \sigma^{(e_i)} + \sigma^{(e_i)} \partial^F &= f_i \operatorname{id}_F, \\ \partial^F \sigma^{(H)} + \sigma^{(H)} \partial^F &= - \sum_{\substack{I+J=H \\ I, J \neq 0}} \sigma^{(I)} \sigma^{(J)} \quad (|H| \geq 2). \end{aligned}$$

The first nontrivial maps are therefore ordinary null-homotopies for multiplication by the regular-sequence elements. The higher maps record the failure of those null-homotopies to satisfy the next coherence relations strictly.

Example 3.3. Let

$$R = k[a, b, c], \quad S = R/(a^2, b^2), \quad M = k,$$

and take the Koszul resolution

$$F = \operatorname{Kos}^R(a, b, c).$$

If e_1, e_2, e_3 are the exterior generators, define

$$\sigma^{(1,0)} = a e_1 \wedge (-), \quad \sigma^{(0,1)} = b e_2 \wedge (-).$$

Then

$$\partial^F \sigma^{(1,0)} + \sigma^{(1,0)} \partial^F = a^2 \text{id}_F,$$

and similarly for b^2 . Since $e_1^2 = e_2^2 = 0$ and $e_1 e_2 = -e_2 e_1$, all higher maps $\sigma^{(H)}$ with $|H| \geq 2$ may be taken to be zero.

3.3 Resolution transfer

Let

$$\Gamma = \Gamma_R(y_1, \dots, y_c)$$

be the divided-power coalgebra on generators y_i of degree 2. Its basis consists of

$$y^{(H)} = y_1^{(h_1)} \cdots y_c^{(h_c)}, \quad H \in \mathbb{N}^c,$$

with comultiplication

$$\Delta(y^{(H)}) = \sum_{I+J=H} y^{(I)} \otimes y^{(J)}.$$

The dual polynomial variables act on Γ by contraction.

Theorem 3.4 (Shamash–Eisenbud resolution transfer). *Let $S = R/(f_1, \dots, f_c)$ be a quotient by an R -regular sequence, and let $F \rightarrow M$ be an R -free resolution of an S -module M equipped with a system of higher homotopies $\{\sigma^{(H)}\}$. Then the graded S -module*

$$S \otimes_R \Gamma \otimes_R F$$

admits a differential obtained by combining the contraction operators on Γ with the maps $\sigma^{(H)}$, and the resulting complex is an S -free resolution of M .

Concretely, if t^H denotes the contraction operator dual to $y^{(H)}$, the differential has the schematic form

$$d = \sum_{H \in \mathbb{N}^c} t^H \otimes \sigma^{(H)}.$$

The higher-homotopy identities are exactly what is needed for $d^2 = 0$.

3.4 Koszul duality and the Priddy coalgebra

Definition 3.5 (Koszul DG algebra). Let A be an augmented DG k -algebra. It is *Koszul* if the residue field k admits a linear semifree resolution over A .

Polynomial rings, exterior algebras, and quadratic monomial algebras provide basic examples. For a Koszul algebra A , the linear resolution of k can be encoded by a coalgebra C , called

the *Priddy coalgebra*, together with a twisting morphism

$$\tau: C \longrightarrow A.$$

In the DG setting relevant here, C also carries a degree -2 curvature map

$$h: C \longrightarrow R.$$

If

$$\eta: R \longrightarrow A$$

is the unit map, the twisting morphism satisfies the corresponding curved Maurer–Cartan equation, whose right-hand side is $\eta \circ h$. The corresponding twisted tensor product

$$A \otimes_{\tau} C$$

is a resolution of k .

Example 3.6 (Polynomial rings). Let

$$A = k[a_1, \dots, a_n].$$

Then

$$C = \bigwedge_k (e_1, \dots, e_n)$$

with

$$\Delta(e_i) = 1 \otimes e_i + e_i \otimes 1$$

and the usual shuffle comultiplication on higher wedges. Define

$$\tau(e_i) = a_i,$$

and let τ vanish on wedge degree different from 1. The twisted tensor product

$$A \otimes_{\tau} C$$

is precisely the Koszul complex

$$\mathrm{Kos}^A(a_1, \dots, a_n).$$

Example 3.7 (The Koszul complex of a sequence). Let

$$A = \mathrm{Kos}^R(f_1, \dots, f_c).$$

The relevant coalgebra is the divided-power coalgebra

$$C = \Gamma_R(y_1, \dots, y_c),$$

with

$$\Delta(y^{(H)}) = \sum_{I+J=H} y^{(I)} \otimes y^{(J)}.$$

If e_i is the i th Koszul generator, the twisting morphism is determined in lowest degree by

$$\tau(y_i) = e_i.$$

The curvature is determined on the degree-2 cogenerators by

$$h(y_i) = d_A(e_i) = f_i,$$

and is zero on the remaining divided-power basis elements. Under this identification, a classical system of higher homotopies becomes a twisting morphism

$$\sigma: C \longrightarrow \text{End}_R(F).$$

Evaluating the curved Maurer–Cartan equation on the primitive element y_i gives

$$\partial^F \sigma(y_i) + \sigma(y_i) \partial^F = f_i \text{id}_F,$$

which is precisely the second condition in the classical Eisenbud–Shamash definition. Thus the curvature term records the differentials of the degree-1 generators of A , while the remaining components of the Maurer–Cartan equation recover the higher coherence relations.

3.5 Koszul homomorphisms and universal resolution transfer

The coalgebraic viewpoint no longer depends on the quotient being a complete intersection.

Theorem 3.8 (Resolution transfer for Koszul homomorphisms). *Let*

$$\varphi: R \longrightarrow S$$

be a Koszul homomorphism, let M be an S -module, and let C be the Priddy coalgebra associated with φ . If

$$F \xrightarrow{\sim} M$$

is a free R -resolution equipped with a system of higher homotopies

$$\sigma: C \longrightarrow \operatorname{End}_R(F),$$

then the twisted complex built on

$$S \otimes_R C \otimes_R F$$

is an S -free resolution of M .

The classical Shamash–Eisenbud construction is recovered when φ is the quotient by a regular sequence. The gain is that the same formalism applies to broader Koszul homomorphisms, including quasi-complete-intersection phenomena.

3.6 Exact zero-divisors

Definition 3.9 (Exact pair of zero-divisors). Let R be a commutative Noetherian ring. Elements $a, b \in R$ form an *exact pair of zero-divisors* if

$$\operatorname{Ann}_R(a) = (b), \quad \operatorname{Ann}_R(b) = (a).$$

Let $S = R/(a)$. For an exact pair (a, b) , the homomorphism

$$R \longrightarrow S$$

is Koszul. The associated Priddy coalgebra has the form

$$C = \Gamma_R(y) \otimes_R \bigwedge_R(x),$$

where

$$|y| = 2, \quad |x| = 3,$$

and

$$\begin{aligned} \Delta(y^{(n)}) &= \sum_{i=0}^n y^{(i)} \otimes y^{(n-i)}, \\ \Delta(x) &= 1 \otimes x + x \otimes 1. \end{aligned}$$

The natural basis is given by the elements

$$y^{(n)} \quad \text{and} \quad xy^{(n)}.$$

Here the degree-2 cogenerator carries the curvature

$$h(y) = a,$$

corresponding to the differential of the degree-1 generator in the Koszul-type DG algebra associated with $R \rightarrow S$; the curvature vanishes on the other cogenerators.

Let F be an R -free resolution of an S -module. A system of higher homotopies can then be written as maps

$$\alpha^n, \beta^n: F \longrightarrow F,$$

where

$$|\alpha^n| = 2n - 1, \quad |\beta^n| = 2n + 2.$$

The twisting-morphism equation becomes the following recursive system:

$$\begin{aligned} \alpha^0 &= \partial^F, \\ \partial^F \alpha^1 + \alpha^1 \partial^F &= a \operatorname{id}_F, \\ \partial^F \alpha^n + \alpha^n \partial^F &= - \sum_{i=1}^{n-1} \alpha^i \alpha^{n-i} \quad (n \geq 2), \\ \partial^F \beta^n - \beta^n \partial^F &= (n+1)b \alpha^{n+1} - \sum_{i=1}^{n-1} (\beta^i \alpha^{n-i} + \alpha^i \beta^{n-i}). \end{aligned}$$

The last family is the extra coherence relation forced by the degree-3 exterior generator. In this way the coalgebraic definition recovers the explicit higher homotopies attached to exact zero-divisors.

Summary of the Lecture

The main point is that classical systems of higher homotopies are not an isolated collection of recursive formulas. They arise from a single twisting morphism

$$C \longrightarrow \operatorname{End}_R(F),$$

where C is the Koszul-dual coalgebra attached to the ring map. For a complete intersection this gives the Shamash–Eisenbud resolution. For a general Koszul homomorphism, the Priddy coalgebra produces the corresponding universal resolution-transfer mechanism. Exact zero-divisors provide a concrete non-complete-intersection example in which the resulting coherence identities can still be written explicitly.

4 Dalena Vien — Edge Ideals via Independence Polynomials of Graphs

Talk information

Title: *Edge Ideals via Independence Polynomials of Graphs*

Speaker: Dalena Vien

Affiliation: Department of Mathematics, Bryn Mawr College

Conference: URiCA 2026, University of Nebraska–Lincoln

Conference dates: August 29–30, 2026

Joint work:

- (1) Jennifer Biermann, Trung Chau, Selvi Kara, Augustine O’Keefe, Joseph Skelton, and Gabriel Sosa Castillo;
- (2) Takayuki Hibi and Selvi Kara.

4.1 Introduction and Motivation

The talk begins with a general translation problem between combinatorics and commutative algebra:

$$\text{graph } G \longleftrightarrow \text{edge ideal } I(G).$$

The guiding question is:

Question 4.1. What can combinatorics tell us about algebra? More specifically, what combinatorial data attached to a graph G determines algebraic information about $I(G)$, and how does the algebraic information change when the graph is modified?

The talk focuses on a particularly concrete bridge: independent sets of G determine the independence polynomial $P_G(x)$, while the edge ideal determines a Hilbert series and hence an h -polynomial. The main result identifies the degree of this h -polynomial using the multiplicity of the root $x = -1$ of $P_G(x)$.

4.2 Edge Ideals and Algebraic Invariants

4.2.1 Edge ideals

Let

$$G = (V, E), \quad V = \{x_1, \dots, x_n\},$$

be a finite simple graph and let

$$S = k[x_1, \dots, x_n]$$

be a polynomial ring over a field k .

Definition 4.2 (Edge ideal). The *edge ideal* of G is the squarefree quadratic monomial ideal

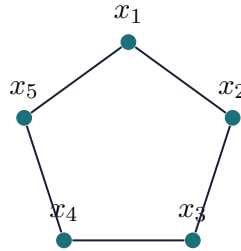
$$I(G) = (x_i x_j : \{x_i, x_j\} \in E(G)) \subseteq S.$$

Example 4.3 (The 5-cycle). For the cycle

$$C_5 : \quad x_1 - x_2 - x_3 - x_4 - x_5 - x_1,$$

one has

$$I(C_5) = (x_1 x_2, x_2 x_3, x_3 x_4, x_4 x_5, x_5 x_1) \subseteq k[x_1, x_2, x_3, x_4, x_5].$$



4.2.2 Minimal graded free resolutions

Let $I \subseteq S$ be a homogeneous ideal. A minimal graded free resolution of S/I has the form

$$0 \longrightarrow \bigoplus_{j \in \mathbb{Z}} S(-j)^{\beta_{p,j}} \longrightarrow \cdots \longrightarrow \bigoplus_{j \in \mathbb{Z}} S(-j)^{\beta_{1,j}} \longrightarrow S \longrightarrow S/I \longrightarrow 0.$$

The graded Betti number $\beta_{i,j}(S/I)$ is the number of copies of $S(-j)$ appearing in homological degree i .

These numbers determine, among other invariants,

$$\text{pdim}(S/I) = \max\{i : \beta_{i,j} \neq 0 \text{ for some } j\},$$

and

$$\text{reg}(S/I) = \max\{j - i : \beta_{i,j} \neq 0\}.$$

Example 4.4 (The path P_3). Consider

$$I(P_3) = (x_1x_2, x_2x_3) \subseteq k[x_1, x_2, x_3].$$

For this displayed ideal, the minimal resolution is

$$0 \longrightarrow S(-3) \xrightarrow{\begin{bmatrix} -x_3 \\ x_1 \end{bmatrix}} S(-2)^2 \xrightarrow{\begin{bmatrix} x_1x_2 & x_2x_3 \end{bmatrix}} S \longrightarrow S/I(P_3) \longrightarrow 0.$$

Thus

$$\beta_{1,2} = 2, \quad \beta_{2,3} = 1.$$

4.3 Independence Complexes and Hochster's Formula

4.3.1 The independence complex

Definition 4.5 (Independence complex). The *independence complex* of G is

$$\Delta(G) = \{F \subseteq V(G) : F \text{ is an independent set of } G\}.$$

The independence number $\alpha(G)$ is the maximum cardinality of an independent set of G . Consequently,

$$\dim \Delta(G) = \alpha(G) - 1.$$

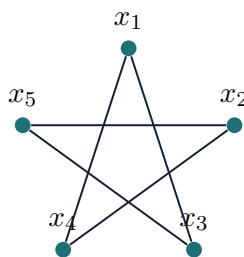
For $G = C_5$, the independent sets are

$$\emptyset, \quad \{x_1\}, \dots, \{x_5\}, \quad \{x_1, x_3\}, \{x_1, x_4\}, \{x_2, x_4\}, \{x_2, x_5\}, \{x_3, x_5\}.$$

Thus

$$\alpha(C_5) = 2,$$

and the 1-skeleton of $\Delta(C_5)$ is again a 5-cycle; with the chosen placement of the vertices, it appears as a pentagram.



4.3.2 Stanley–Reisner translation

The Stanley–Reisner ideal of $\Delta(G)$ is generated by the minimal nonfaces of $\Delta(G)$.

Since the minimal nonfaces of an independence complex are precisely the edges of G ,

$$I_{\Delta(G)} = I(G).$$

This permits topological information about induced subcomplexes of $\Delta(G)$ to control the Betti numbers of the edge ideal.

Theorem 4.6 (Hochster's formula). *For $W \subseteq V(G)$, let $\Delta(G)_W$ denote the induced subcomplex on W .*

Then

$$\beta_{i,j}(S/I_{\Delta(G)}) = \sum_{\substack{W \subseteq V(G) \\ |W|=j}} \dim_k \widetilde{H}_{j-i-1}(\Delta(G)_W; k).$$

Because

$$I_{\Delta(G)} = I(G),$$

this computes the graded Betti numbers of $S/I(G)$ combinatorially.

4.3.3 Fröberg's theorem as a model application

A central example of the combinatorics–algebra correspondence is the following result quoted in the talk.

Theorem 4.7 (Fröberg, 1990). *For a graph G with at least one edge,*

$$\operatorname{reg}(S/I(G)) = 1 \iff \overline{G} \text{ is chordal.}$$

Here chordal means that every cycle of length at least 4 has a chord; equivalently, the graph has no induced cycle of length at least 4.

For example, consider the graph whose edge ideal is

$$I(G) = (x_1x_3, x_3x_5, x_1x_4) \subseteq k[x_1, x_2, x_3, x_4, x_5].$$

Its complement is chordal, hence

$$\operatorname{reg}(S/I(G)) = 1.$$

The broader point is that expressing graded Betti numbers in terms of $\Delta(G)$ gives a powerful method for reading algebraic information from graph-theoretic or topological data.

4.4 Hilbert Series and the h -Polynomial

4.4.1 Hilbert series from the resolution

The Hilbert series of $S/I(G)$ is

$$H_{S/I(G)}(t) = \sum_{d \geq 0} \dim_k(S/I(G))_d t^d.$$

The graded free resolution yields the expression

$$H_{S/I(G)}(t) = \frac{\sum_{i \geq 0} \left(\sum_{j \in \mathbb{Z}} (-1)^i \beta_{i,j} t^j \right)}{(1-t)^n}.$$

This expression need not yet be reduced.

Since

$$\dim(S/I(G)) = \alpha(G),$$

we may write the Hilbert series in reduced form as

$$H_{S/I(G)}(t) = \frac{h_{S/I(G)}(t)}{(1-t)^{\alpha(G)}},$$

where $h_{S/I(G)}(t)$ is the h -polynomial.

For brevity, write

$$h_G(t) = h_{S/I(G)}(t) = \sum_{i \geq 0} h_i(G) t^i.$$

4.4.2 The f -vector and h -vector

If

$$f(\Delta(G)) = (f_{-1}, f_0, \dots, f_{\alpha(G)-1}),$$

then f_{i-1} counts independent sets of cardinality i .

In particular, for C_5 ,

$$f(\Delta(C_5)) = (1, 5, 5).$$

The corresponding h -polynomial is

$$h_{S/I(C_5)}(t) = 1 + 3t + t^2,$$

so its h -vector is

$$(1, 3, 1).$$

Determining the entire h -vector can be difficult. A coarser but still meaningful invariant is

$$\deg h_{S/I(G)}(t) = \max\{i : h_i \neq 0\}.$$

4.4.3 What was known about $\deg h$

If S/I is Cohen–Macaulay, then

$$\operatorname{reg}(S/I) = \deg h_{S/I}(t).$$

Without the Cohen–Macaulay hypothesis, the two invariants may behave independently.

The following prior results motivate the problem.

- **Hibi–Matsuda (2018).** For every pair of integers $r, d \geq 1$, there is a monomial ideal I for which

$$\operatorname{reg}(S/I) = r, \quad \deg h_{S/I}(t) = d.$$

- **Hibi–Matsuda–Van Tuyl (2019).** For every pair $r, d \geq 1$, such a pair is also realized by an edge ideal. Moreover, if G has n vertices, then

$$\operatorname{reg}(S/I(G)) + \deg h_{S/I(G)}(t) \leq n.$$

- **Biermann–Kara–O’Keefe–Skelton–Sosa Castillo (2025).** Explicit formulas for $\deg h$ were obtained for several graph families, including paths, cycles, and complete bipartite graphs.

The goal is to express

$$\deg h_{S/I(G)}(t)$$

entirely in terms of combinatorial data from an arbitrary graph G .

4.5 From Independent Sets to the Degree of the h -Polynomial

4.5.1 Expressing the h -polynomial using $\Delta(G)$

Let g_i denote the number of independent sets of G of size i .

Then

$$g_i = f_{i-1} \quad (0 \leq i \leq \alpha(G)),$$

with

$$g_0 = 1$$

corresponding to the empty set.

The f -vector-to- h -vector relation gives

$$h_{S/I(G)}(t) = \sum_{i=0}^{\alpha(G)} g_i t^i (1-t)^{\alpha(G)-i}. \quad (4.1)$$

In particular,

$$\deg h_{S/I(G)}(t) \leq \alpha(G).$$

4.5.2 The top coefficient and the reduced Euler characteristic

The coefficient in top possible degree satisfies

$$h_{\alpha(G)} = (-1)^{\alpha(G)-1} \tilde{\chi}(\Delta(G)),$$

where

$$\tilde{\chi}(\Delta(G)) = \sum_{i=0}^{\alpha(G)} (-1)^{i-1} f_{i-1} = \sum_{i=0}^{\alpha(G)} (-1)^{i-1} g_i.$$

Thus maximal degree is controlled by an alternating sum of the numbers of independent sets.

4.5.3 The independence polynomial

Definition 4.8 (Independence polynomial). The *independence polynomial* of G is

$$P_G(x) = \sum_{i=0}^{\alpha(G)} g_i x^i.$$

For example,

$$P_{C_5}(x) = 1 + 5x + 5x^2.$$

The previous alternating sum is precisely the special value at -1 :

$$\tilde{\chi}(\Delta(G)) = -P_G(-1).$$

Consequently,

$$h_{\alpha(G)} = (-1)^{\alpha(G)} P_G(-1),$$

and hence

$$\deg h_{S/I(G)}(t) = \alpha(G) \iff P_G(-1) \neq 0. \quad (4.2)$$

4.5.4 Multiplicity of the root -1

When

$$P_G(-1) = 0,$$

write

$$P_G(x) = (1+x)^{M(G)}Q(x), \quad Q(-1) \neq 0,$$

where

Definition 4.9.

$$M(G) = \text{ord}_{x=-1} P_G(x)$$

is the multiplicity of $x = -1$ as a root of the independence polynomial.

The main theorem identifies the exact amount by which the degree drops from its maximal possible value.

Theorem 4.10 (Biermann–Chau–Kara–O’Keefe–Skelton–Sosa–Castillo–Vien). *For every finite simple graph G ,*

$$\deg h_{S/I(G)}(t) = \alpha(G) - M(G).$$

Thus each additional order of vanishing of $P_G(x)$ at $x = -1$ lowers the degree of the h -polynomial by one.

Since

$$\dim(S/I(G)) = \alpha(G),$$

the theorem also yields

$$a(S/I(G)) = \deg h_{S/I(G)}(t) - \dim(S/I(G)) = -M(G).$$

The same change-of-variables argument applies more generally to Stanley–Reisner rings: the degree drop is governed by the order of vanishing at -1 of the face-enumerating polynomial. In particular, the graph statement has a squarefree-monomial, or hypergraph, analogue.

4.5.5 Proof of the degree formula

Proof. Starting from (4.1), factor the expression through the independence polynomial:

$$\begin{aligned} h_{S/I(G)}(t) &= \sum_{i=0}^{\alpha(G)} g_i t^i (1-t)^{\alpha(G)-i} \\ &= (1-t)^{\alpha(G)} P_G\left(\frac{t}{1-t}\right). \end{aligned}$$

Set

$$u = \frac{1}{1-t}, \quad \frac{t}{1-t} = u - 1.$$

Then

$$h_{S/I(G)}(t) = u^{-\alpha(G)} P_G(u - 1).$$

If

$$P_G(x) = (1 + x)^{M(G)} Q(x), \quad Q(-1) \neq 0,$$

then

$$h_{S/I(G)}(t) = u^{M(G)-\alpha(G)} Q(u - 1).$$

Equivalently, with

$$d = \alpha(G) - M(G),$$

we obtain

$$h_{S/I(G)}(t) = (1 - t)^d Q\left(\frac{t}{1 - t}\right).$$

Because

$$\deg P_G = \alpha(G),$$

one has

$$\deg Q = d.$$

Writing

$$Q(x) = \sum_{i=0}^d q_i x^i,$$

gives

$$h_{S/I(G)}(t) = \sum_{i=0}^d q_i t^i (1 - t)^{d-i}.$$

The coefficient of t^d is

$$\sum_{i=0}^d q_i (-1)^{d-i} = (-1)^d Q(-1),$$

which is nonzero.

Hence

$$\deg h_{S/I(G)}(t) = d = \alpha(G) - M(G).$$

□

4.5.6 Examples

Example 4.11 (The cycle C_5). We have

$$P_{C_5}(x) = 1 + 5x + 5x^2.$$

Since

$$P_{C_5}(-1) = 1,$$

one has

$$\alpha(C_5) = 2, \quad M(C_5) = 0.$$

Therefore

$$\deg h_{S/I(C_5)}(t) = 2.$$

Example 4.12 (The path P_4). A direct count gives

$$P_{P_4}(x) = 1 + 4x + 3x^2 = (1 + x)(1 + 3x).$$

Thus

$$\alpha(P_4) = 2, \quad M(P_4) = 1,$$

and Theorem 4.10 gives

$$\deg h_{S/I(P_4)}(t) = 1.$$

4.6 Pseudo-Gorenstein and Pseudo-Gorenstein*

4.6.1 Related notions

Let A be a standard graded Cohen–Macaulay k -algebra with h -vector

$$(h_0, h_1, \dots, h_s).$$

If A is Gorenstein, then its h -vector is symmetric and in particular $h_s = 1$. The pseudo-Gorenstein condition isolates this last numerical feature: in the Cohen–Macaulay setting, A is called *pseudo-Gorenstein* when its last nonzero h -coefficient is 1.

For edge ideals, Hibi–Kara–Vien consider the corresponding graph condition and single out the extremal case in which the h -polynomial has the largest possible degree.

Definition 4.13 (Pseudo-Gorenstein* graph; Hibi–Kara–Vien). A graph G is called *pseudo-Gorenstein** if

$$\deg h_G(t) = \alpha(G) \quad \text{and} \quad h_{\alpha(G)}(G) = 1.$$

Equivalently, the a -invariant of $S/I(G)$ is 0 and the top h -coefficient is 1.

Using

$$h_{\alpha(G)} = (-1)^{\alpha(G)} P_G(-1),$$

we obtain the following criterion.

Proposition 4.14 (Hibi–Kara–Vien criterion). *A graph G is pseudo-Gorenstein* if and only if*

$$P_G(-1) = (-1)^{\alpha(G)}.$$

4.6.2 Paths and cycles

The independence polynomial satisfies the standard vertex recursion

$$P_G(x) = P_{G-v}(x) + xP_{G-N[v]}(x), \quad (4.3)$$

where $N[v]$ is the closed neighborhood of v .

In particular,

$$P_{P_n}(x) = P_{P_{n-1}}(x) + xP_{P_{n-2}}(x),$$

and

$$P_{C_n}(x) = P_{P_{n-1}}(x) + xP_{P_{n-3}}(x).$$

At $x = -1$, the path values become periodic. The beginning of the sequence is

n	1	2	3	4	5	6	7
$P_{P_n}(-1)$	0	-1	-1	0	1	1	0

and the cycle values are periodic as well.

Thus Proposition 4.14 reduces the classification to congruence conditions.

The resulting classifications are:

$$\begin{aligned} C_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 2, 5, 10 \pmod{12}, \\ P_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0, 2, 9, 11 \pmod{12}. \end{aligned}$$

4.7 Graph Suspensions

4.7.1 C -suspensions

Definition 4.15 (C -suspension). Let

$$\emptyset \neq C \subseteq V(G).$$

The C -suspension

$$G(C)$$

is obtained from G by adjoining a new vertex z and joining z to every vertex of C .

Independent sets of $G(C)$ split according to whether they contain z .

This gives

$$P_{G(C)}(x) = P_G(x) + xP_{G-C}(x). \quad (4.4)$$

The point of this identity is that both $P_G(-1)$ and the multiplicity $M(G)$ can be tracked under suspension.

4.7.2 Suspending over a vertex cover

Definition 4.16 (Vertex cover). A set

$$C \subseteq V(G)$$

is a *vertex cover* if every edge of G is incident to at least one vertex of C .

If C is a vertex cover and

$$S_0 = V(G) \setminus C, \quad s = |S_0|,$$

then S_0 is independent, so $G - C$ is edgeless.

Hence

$$P_{G-C}(x) = (1 + x)^s,$$

and (4.4) becomes

$$P_{G(C)}(x) = P_G(x) + x(1 + x)^s. \quad (4.5)$$

In particular, if

$$s \geq 1,$$

then

$$P_{G(C)}(-1) = P_G(-1).$$

Theorem 4.17 (Hibi–Kara–Vien). *Let C be a vertex cover of G such that*

$$0 < |V(G) \setminus C| < \alpha(G).$$

Then

$$G \text{ is pseudo-Gorenstein}^* \iff G(C) \text{ is pseudo-Gorenstein}^*.$$

More precisely, if

$$H = G(C), \quad s = |V(G) \setminus C|, \quad \alpha = \alpha(H) = \alpha(G),$$

then

$$P_H(-1) = P_G(-1)$$

and

$$h_{S[z]/I(H)}(t) = h_{S/I(G)}(t) + t(1-t)^{\alpha-s-1}. \quad (4.6)$$

Under the hypotheses of Theorem 4.17, this implies

$$h_\alpha(H) = h_\alpha(G).$$

4.7.3 Extremal values of s

The preservation theorem excludes the two extremal cases

$$s = 0 \quad \text{and} \quad s = \alpha(G).$$

Suppose first that

$$s = \alpha(G).$$

Then

$$\alpha(H) = \alpha(G) + 1$$

and

$$h_{S[z]/I(H)}(t) = (1-t)h_{S/I(G)}(t) + t. \quad (4.7)$$

If G is pseudo-Gorenstein*, so that

$$h_{\alpha(G)}(G) = 1,$$

then the new leading coefficient is

$$h_{\alpha(H)}(H) = -1.$$

Thus this extremal suspension does not preserve the pseudo-Gorenstein* property.

For example, start with

$$G = C_5$$

and chooses a vertex cover whose complement has size

$$s = \alpha(C_5) = 2.$$

After adjoining the suspension vertex z , the graph H has an independent set of size 3, namely the two vertices outside the cover together with z , so

$$\alpha(H) = 3.$$

4.7.4 Full suspensions

A *full suspension* corresponds to

$$C = V(G),$$

hence

$$s = 0.$$

Full suspension does not preserve pseudo-Gorenstein* in general.

Writing

$$\widehat{G}$$

for the full suspension of G , the following classification is given for paths and cycles.

Theorem 4.18 (Hibi–Kara–Vien). *For full suspensions of cycles and paths,*

$$\begin{aligned} \widehat{C}_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0 \pmod{12}, \\ \widehat{P}_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 10 \pmod{12}. \end{aligned}$$

For comparison, the unsuspended families satisfy

$$\begin{aligned} C_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 2, 5, 10 \pmod{12}, \\ P_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0, 2, 9, 11 \pmod{12}. \end{aligned}$$

4.8 Key Takeaways and Future Directions

The discussion closes by returning to the original combinatorics–algebra dictionary.

The chain of ideas is

$$\text{graph } G \longrightarrow P_G(x) \longleftrightarrow h_{S/I(G)}(t) \longleftarrow I(G),$$

with the precise degree formula

$$\deg h_{S/I(G)}(t) = \alpha(G) - M(G).$$

In particular, the special value $P_G(-1)$ simultaneously records:

- combinatorial data of the graph G ;
- topological data of the independence complex $\Delta(G)$ through its reduced Euler characteristic;
- algebraic data encoded by the Hilbert series and h -polynomial of the edge ideal.

Natural directions suggested by these results include:

- (i) understand $M(G)$ under other graph operations;
- (ii) find a graph-theoretic interpretation of

$$M(G) = 0, \quad \text{equivalently} \quad P_G(-1) \neq 0;$$

- (iii) classify graph families according to the behavior of $P_G(x)$ at $x = -1$.

Selected Attributions

The principal results used in this talk are attributed as follows.

- Fröberg (1990): characterization of edge ideals with linear resolution via chordality of the complement.
- Hibi–Matsuda (2018): realizability of arbitrary positive pairs $(\text{reg}, \deg h)$ for monomial ideals.
- Hibi–Matsuda–Van Tuyl (2019): realizability for edge ideals and the inequality

$$\text{reg}(S/I(G)) + \deg h_{S/I(G)}(t) \leq n.$$

- Biermann–Kara–O’Keefe–Skelton–Sosa Castillo (2025): formulas for $\deg h$ for several graph families.
- Ene–Herzog–Hibi–Saeedi Madani: the pseudo-Gorenstein condition for standard graded Cohen–Macaulay algebras.
- Biermann–Chau–Kara–O’Keefe–Skelton–Sosa Castillo–Vien: the degree formula in Theorem 4.10.
- Hibi–Kara–Vien: the pseudo-Gorenstein* criterion and the suspension results described above.

Talk information

Title: *The graded Möbius algebra of a matroid*

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5 Sean Grate – The graded Möbius algebra of a matroid

5.1 Free resolutions and numerical invariants

The algebraic background begins with the information encoded by a minimal graded free resolution. Let

$$S = \mathbb{K}[x_1, \dots, x_n]$$

be a standard graded polynomial ring, let $I \subseteq S$ be a homogeneous ideal, and set

$$R = S/I.$$

Its minimal graded free resolution over S has the form

$$0 \longleftarrow R \longleftarrow F_0 \longleftarrow F_1 \longleftarrow \cdots \longleftarrow F_p \longleftarrow 0, \quad p = \text{pdim}_S R \leq n,$$

where

$$F_i \cong \bigoplus_j S(-i-j)^{\beta_{i,i+j}^S(R)}.$$

The integers

$$\beta_{i,i+j}^S(R)$$

are the graded Betti numbers, while

$$\beta_i^S(R) = \sum_j \beta_{i,i+j}^S(R)$$

are the total Betti numbers.

The usual Betti-table convention places $\beta_{i,i+j}$ in column i and row j . With this convention, the Castelnuovo–Mumford regularity is the largest nonzero row index:

$$\text{reg}_S(R) = \max \left\{ j : \beta_{i,i+j}^S(R) \neq 0 \text{ for some } i \right\},$$

whereas the projective dimension is the largest nonzero column index:

$$\text{pdim}_S(R) = \max \left\{ i : \beta_{i,i+j}^S(R) \neq 0 \text{ for some } j \right\}.$$

Example 5.1. Consider the Betti table

	0	1	2	3
0	1	.	.	.
1	.	2	.	.
2	.	1	1	.
3	.	.	2	.
4	.	.	.	1

Then

$$\operatorname{reg}_S(R) = 4, \quad \operatorname{pdim}_S(R) = 3.$$

If

$$S = \mathbb{K}[x, y, z],$$

the alternating sum of the free modules gives

$$H_R(t) = \frac{1 - (2t^2 + t^3) + (t^4 + 2t^5) - t^7}{(1 - t)^3}.$$

Hence

$$H_R(t) = 1 + 3t + 4t^2 + \cdots.$$

Thus the same resolution simultaneously records homological invariants and the Hilbert series.

5.2 Koszul algebras

Let

$$R = \bigoplus_{i \geq 0} R_i$$

be a standard graded $\mathbb{K}$ -algebra and let

$$\mathfrak{m} = \bigoplus_{i \geq 1} R_i.$$

Definition 5.2. The algebra R is *Koszul* if the residue field

$$R/\mathfrak{m} \cong \mathbb{K}$$

has a linear minimal free resolution as an R -module.

Example 5.3. For

$$R = \mathbb{K}[x, y, z],$$

the residue field has the Koszul resolution

$$0 \longleftarrow \mathbb{K} \longleftarrow R \longleftarrow R(-1)^3 \longleftarrow R(-2)^3 \longleftarrow R(-3) \longleftarrow 0.$$

With the indicated bases, the differentials are

$$d_1 = \begin{pmatrix} x & y & z \end{pmatrix}, \quad d_2 = \begin{pmatrix} -y & -z & 0 \\ x & 0 & -z \\ 0 & x & y \end{pmatrix}, \quad d_3 = \begin{pmatrix} z \\ -y \\ x \end{pmatrix}.$$

Every differential is represented by linear forms. The Betti table therefore consists of a single row:

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 1 & 3 & 3 & 1 \end{array}.$$

5.2.1 When does quadratic imply Koszul?

A Koszul algebra must be quadratic, but the converse fails. Even among Gorenstein algebras presented by quadrics, the answer depends sharply on regularity and codimension.

Theorem 5.4 (Conca–Rossi–Valla, 2001). *Let R be a graded Gorenstein $\mathbb{K}$ -algebra presented by quadrics. If*

$$\operatorname{reg}(R) \leq 2,$$

or if

$$\operatorname{reg}(R) = 3 \quad \text{and} \quad \operatorname{codim}(R) \leq 4,$$

then R is Koszul.

Theorem 5.5 (Mastroeni–Schenck–Stillman, 2021/2023). *For every pair of integers*

$$c \geq r + 2 \geq 6,$$

there exists a graded quadratic Gorenstein $\mathbb{K}$ -algebra R satisfying

$$\operatorname{reg}(R) = r, \quad \operatorname{codim}(R) = c,$$

which is not Koszul.

There is also no universal finite test for Koszulness obtained by inspecting only a fixed initial segment of the resolution of the residue field.

Theorem 5.6 (Roos, 1993). *Koszulness of a standard graded algebra cannot, in general, be decided from finitely many prescribed steps of the minimal free resolution of its residue field.*

5.2.2 Restrictions imposed by Koszulness

The Poincaré series of R is

$$P_R(t) = \sum_{i \geq 0} \beta_i^R(\mathbb{K}) t^i,$$

the generating function for the total Betti numbers in the minimal R -free resolution of $\mathbb{K}$.

Proposition 5.7. *Let $R = S/I$ be a standard graded Koszul algebra. Then:*

(i) *I is generated by quadrics;*

(ii)

$$\operatorname{reg}_S(R) \leq \operatorname{pdim}_S(R);$$

(iii)

$$\beta_{i,i+j}^S(R) = 0 \quad \text{for } j > i;$$

(iv) *if c is the number of quadratic minimal generators of I , then*

$$\beta_{i,2i}^S(R) \leq \binom{c}{i};$$

(v) *the Poincaré and Hilbert series satisfy*

$$P_R(t)H_R(-t) = 1.$$

Two standard methods for proving Koszulness are particularly relevant here.

First, one may construct a *Koszul filtration*: a family of ideals generated by linear forms, containing 0 and $\mathfrak{m}$, and recursively stable under suitable colon operations. Such filtrations can be effective, although they are often large.

Second, one may prove that R is G -quadratic, meaning that its defining ideal admits a quadratic Gröbner basis. In that case the corresponding quadratic monomial initial algebra is Koszul, and the usual deformation argument yields the Koszulness of R .

5.3 Matroids and geometric lattices

5.3.1 A running example: the Fano matroid

The basic combinatorial example is the projective plane

$$PG(2, 2) = F_7.$$

It is represented over $\mathbb{F}_2$ by the columns of

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix},$$

which we label

$$p_1, \dots, p_7.$$

Its seven rank-two flats are

$$127, \quad 136, \quad 145, \quad 235, \quad 246, \quad 347, \quad 567,$$

where, for example,

$$127 = \{p_1, p_2, p_7\}.$$

Definition 5.8. A *matroid* is a pair

$$M = (E, \mathcal{I}),$$

where E is a finite ground set and

$$\mathcal{I} \subseteq 2^E$$

is the collection of independent sets, satisfying:

(i)

$$\emptyset \in \mathcal{I};$$

(ii) if

$$I' \subseteq I \quad \text{and} \quad I \in \mathcal{I},$$

then

$$I' \in \mathcal{I};$$

(iii) if

$$I, J \in \mathcal{I} \quad \text{and} \quad |I| < |J|,$$

then there exists

$$e \in J \setminus I$$

such that

$$I \cup \{e\} \in \mathcal{I}.$$

Linear independence in a vector space gives the prototypical example. Other standard sources include cycle matroids of graphs and matroids associated to hyperplane arrangements; Boolean matroids give the extremal free example.

Maximal independent sets are the *bases*. Minimal dependent sets are the *circuits*.

For $F \subseteq E$, the rank is

$$\text{rank}(F) = \max \{|I| : I \subseteq F, I \in \mathcal{I}\},$$

and the closure is

$$\text{cl}(F) = \{e \in E : \text{rank}(F \cup \{e\}) = \text{rank}(F)\}.$$

A set F is a *flat* if

$$\text{cl}(F) = F.$$

Example 5.9. For F_7 , one has

Set	Type and rank	Closure
$\{p_2, p_3, p_4\}$	basis, rank 3	$\{p_1, \dots, p_7\}$
$\{p_2, p_3\}$	independent, rank 2	$\{p_2, p_3, p_5\}$
$\{p_1, p_2, p_3, p_4\}$	circuit, rank 3	$\{p_1, \dots, p_7\}$
$\{p_3, p_4, p_7\}$	dependent flat, rank 2	$\{p_3, p_4, p_7\}$

5.3.2 Geometric lattices

Definition 5.10. A *lattice* is a poset in which any two elements F, G have a join

$$F \vee G$$

and a meet

$$F \wedge G.$$

A finite ranked lattice is called *geometric* if it is atomic and upper semimodular. In rank-function language, upper semimodularity is expressed by

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) \leq \text{rank}(F) + \text{rank}(G).$$

Theorem 5.11. *A finite lattice is geometric if and only if it is isomorphic to the lattice of flats of a matroid.*

For the Fano matroid, the lattice of flats has four ranks. Writing only the labels of the flats, its layers are

$$\begin{aligned}\text{rank } 3 & : 1234567, \\ \text{rank } 2 & : 127, 136, 145, 235, 246, 347, 567, \\ \text{rank } 1 & : 1, 2, 3, 4, 5, 6, 7, \\ \text{rank } 0 & : \emptyset.\end{aligned}$$

Thus the Whitney numbers of the second kind are

$$(1, 7, 7, 1).$$

5.3.3 Comparing the ranks

Let $\mathcal{L}$ be a geometric lattice of rank $n > 1$, and set

$$\mathcal{L}^k = \{F \in \mathcal{L} : \text{rank}(F) = k\}, \quad W_k = |\mathcal{L}^k|.$$

Theorem 5.12 (Greene, 1970). *For*

$$k = 2, \dots, n-1,$$

one has

$$W_1 \leq W_k.$$

Moreover,

$$W_1 = W_k$$

if and only if

$$k = n-1$$

and $\mathcal{L}$ is modular.

Here modularity means that for every

$$F, G \in \mathcal{L},$$

one has

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) = \text{rank}(F) + \text{rank}(G).$$

The stronger top-heavy statement compares all ranks symmetrically placed about the middle.

Theorem 5.13 (Huh–Wang, 2017; Braden–Huh–Matherne–Proudfoot–Wang, 2020).
Let $\mathcal{L}$ be a geometric lattice of rank r . Whenever

$$k \leq j \leq r - k,$$

there exists an injective map

$$\iota : \mathcal{L}^k \hookrightarrow \mathcal{L}^j.$$

In particular,

$$W_k \leq W_j \quad (k \leq j \leq r - k).$$

5.4 The graded Möbius algebra

Let M be a simple matroid with lattice of flats

$$\mathcal{L}(M).$$

Definition 5.14. The *graded Möbius algebra* of M is the standard graded $\mathbb{K}$ -algebra

$$B_M = \bigoplus_{F \in \mathcal{L}(M)} \mathbb{K}x_F,$$

where

$$\deg(x_F) = \text{rank}(F),$$

and multiplication is defined by

$$x_F x_G = \begin{cases} x_{F \vee G}, & \text{rank}(F \vee G) = \text{rank}(F) + \text{rank}(G), \\ 0, & \text{otherwise.} \end{cases}$$

The construction turns the rank data of the lattice directly into the Hilbert function.

Proposition 5.15. *Let M be a simple matroid.*

(i) *For each i ,*

$$\dim_{\mathbb{K}}(B_M)_i = |\{F \in \mathcal{L}(M) : \text{rank}(F) = i\}|.$$

Thus the Hilbert function of B_M is the rank-generating function of $\mathcal{L}(M)$.

(ii)

$$\text{reg}(B_M) = \text{rank}(M) = \text{rank } \mathcal{L}(M).$$

(iii) *If $E = E(M)$ and B_M is viewed as a quotient of the polynomial ring on one variable*

for each element of E , then

$$\text{pdim}(B_M) = |E|.$$

For

$$M = F_7,$$

the Hilbert function is

$$1, 7, 7, 1.$$

Consequently,

$$\text{reg}(B_{F_7}) = 3, \quad \text{pdim}(B_{F_7}) = 7.$$

5.4.1 A presentation by circuits

For a subset

$$A \subseteq E(M),$$

write

$$y_A = \prod_{i \in A} y_i.$$

If

$$\mathcal{C}(M)$$

denotes the set of circuits of M , then a convenient presentation is the following.

Theorem 5.16 (LaClair–Mastroeni–McCullough–Peeva). *Let M be a simple matroid. Then*

$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{\left(y_i^2 : i \in E(M)\right) + \left(y_{C \setminus \{i\}} - y_{C \setminus \{j\}} : C \in \mathcal{C}(M), i, j \in C\right)}.$$

This presentation makes the connection with chordality transparent: a circuit of size d contributes relations of degree $d - 1$, and the possibility of reducing large circuits through smaller ones controls whether the defining ideal is quadratic or has a quadratic Gröbner basis.

5.4.2 The graphic case and the Orlik–Solomon parallel

When

$$M = M(G)$$

is the cycle matroid of a graph G , the graded Möbius algebra exhibits a close parallel with the Orlik–Solomon algebra A_M .

The relationships relevant here may be summarized schematically as

$$\begin{array}{ccc}
 A_M \text{ has a quadratic Gröbner basis} & \iff & M \text{ is supersolvable} \\
 \Downarrow & & \Downarrow \\
 A_M \text{ is Koszul} & & \\
 \Downarrow & & \\
 A_M \text{ is quadratic} & \iff & G \text{ is chordal,}
 \end{array}$$

while on the graded Möbius side one has

$$\begin{array}{ccc}
 B_M \text{ has a quadratic Gröbner basis} & \iff & M \text{ is strongly } T\text{-chordal} \\
 \Downarrow & & \Downarrow \\
 B_M \text{ is Koszul} & \iff & G \text{ is strongly chordal} \\
 \Downarrow & & \\
 B_M \text{ is quadratic} & \iff & G \text{ is chordal.}
 \end{array}$$

In particular, for graphic matroids, Koszulness of the graded Möbius algebra is characterized by strong chordality of the underlying graph.

5.5 Modularity, Gorensteinness, and projective geometries

A basic structural result identifies the Gorenstein graded Möbius algebras.

Theorem 5.17 (Greene, 1970; Maeno–Numata, 2016). *For a simple matroid M , the following are equivalent:*

- (i) *the lattice $\mathcal{L}(M)$ is modular;*
- (ii) *the rank-generating function of $\mathcal{L}(M)$ is symmetric;*
- (iii) *the Hilbert function of B_M is symmetric;*
- (iv) *B_M is Gorenstein.*

The inverse-system description is also useful. If

$$\mathcal{B}(M)$$

is the set of bases of M , then, with the polynomial variables acting by contraction on the dual polynomial ring, B_M may be realized as

$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{\text{Ann}\left(\sum_{B \in \mathcal{B}(M)} y_B\right)}.$$

The principal new result highlighted in the talk concerns finite projective geometries.

Theorem 5.18 (Grate–McCullough). *If M is a finite projective geometry, then*

$$B_M$$

is Koszul.

More concretely, for

$$M = PG(d, q),$$

the argument fits into the chain

$$PG(d, q) \implies M \text{ strongly } T\text{-chordal} \iff B_M \text{ is } G\text{-quadratic} \implies B_M \text{ is Koszul.}$$

The same projective geometries are chordal, which also yields the weaker conclusion that

$$B_M$$

is quadratic.

Since projective geometries have modular lattices of flats, their graded Möbius algebras are simultaneously Gorenstein and Koszul. Thus this family provides a natural class in which Gorensteinness and Koszulness coexist for combinatorial reasons.

5.6 Affine geometries

Finite affine geometries arise from projective geometries by deleting the points on a hyperplane at infinity.

The corresponding graded Möbius algebras behave differently from the projective case.

For

$$M = AG(d, q),$$

the results discussed in the talk give substantial evidence toward a general Koszulness statement.

In particular, for

$$q \geq 3,$$

the algebra B_M is quadratic. When q is odd, the affine geometry is chordal, and quadraticity also follows from the general chordality criterion.

The smallest instructive case is

$$AG(2, 3).$$

Its graded Möbius algebra is Koszul, although its natural presentation does not admit a quadratic Gröbner basis. Thus Koszulness and G -quadraticity already diverge within this family.

Conjecture 5.19 (Grate–McCullough). *Let*

$$M = AG(d, q), \quad d \geq 2, \quad q \geq 3.$$

Then the graded Möbius algebra

$$B_M$$

is Koszul.

The example $AG(2, 3)$ is useful for two reasons. It supports the general Koszulness conjecture while showing that a proof cannot simply proceed by establishing G -quadraticity in the natural circuit presentation.

5.7 Further directions

The central open-ended problem is to determine which matroids have Koszul graded Möbius algebras.

The families discussed in the talk can be summarized as follows:

Matroid family	Status of B_M
Cycle matroids $M(G)$	Koszul exactly when G is strongly chordal
Projective geometries $PG(n, q)$, $n \geq 2$	Koszul
Affine geometries $AG(n, q)$, $n \geq 2$, $q \geq 3$	Conjecturally Koszul; known to be quadratic
Sylvester matroids	Not Koszul in general
Lattice-path matroids	Open
Transversal matroids	Open
Sparse paving and paving matroids	Open
Dowling geometries	Open

These questions sit at the intersection of matroid structure and homological algebra. The graded Möbius algebra remembers the Whitney numbers through its Hilbert function, while its defining equations, Gröbner bases, and free resolution distinguish considerably finer combinatorial behavior.

The projective and affine families indicate that structural notions of matroid chordality should continue to play a central role in understanding Koszulness.

6 Jelena Mojsilović — Residual Intersections and Their Connections to Linkage

6.1 Background and motivation

Residual intersection theory was introduced by Artin and Nagata as a natural generalization of classical linkage. Geometrically, if X is an algebraic variety and $Y \subseteq Z \subseteq X$ are closed subschemes, then the residual intersection of Y in Z is another closed subscheme W such that $W \cup Y = Z$. The algebraic theory studies the corresponding ideal-theoretic construction.

6.2 Basic definitions

Definition 6.1. Let R be a Noetherian ring and let $I \subseteq R$ be an ideal of height g . Let $\mathfrak{a} = (a_1, \dots, a_s) \subseteq I$, with $s \geq g$, and set

$$J = \mathfrak{a} : I.$$

Then J is called an *s-residual intersection* of I if $\text{ht}(J) \geq s$. It is called a *geometric s-residual intersection* if, in addition,

$$\text{ht}(I + J) > s.$$

The case $s = g$ is the one most closely related to classical linkage.

Example 6.2. Let I and J be prime ideals in a Noetherian ring R , with $\text{ht}(I) = g$, $I \not\subseteq J$, and suppose that $I \cap J$ is generated by s elements. If $\mathfrak{a} = I \cap J$, then $J = \mathfrak{a} : I$ is a geometric s -residual intersection of I .

Remark 6.3. Assume that R is local Gorenstein and that I is an unmixed ideal of height g . Then (geometric) linkage is the same as a (geometric) g -residual intersection.

6.3 Linkage and the Peskine–Szpiro theorem

A fundamental theorem of Peskine and Szpiro explains why linkage is so useful in the Cohen–Macaulay setting.

Theorem 6.4 (Peskine–Szpiro). *Let R be a local Gorenstein ring, and suppose that ideals I and J are directly linked by a complete intersection $\mathfrak{a} \subseteq I \cap J$; that is,*

$$J = \mathfrak{a} : I, \quad I = \mathfrak{a} : J.$$

Then the following hold.

- (a) R/I is Cohen–Macaulay if and only if R/J is Cohen–Macaulay.
- (b) The canonical modules are given by

$$\omega_{R/I} \cong J/\mathfrak{a}, \quad \omega_{R/J} \cong I/\mathfrak{a}.$$

This leads naturally to the following question.

Question 6.5. To what extent do the conclusions of the Peskine–Szpiro theorem remain valid for s -residual intersections when $s \geq g + 1$?

In general the answer is negative. Nevertheless, strong positive results are known in several important classes, including complete intersections, perfect ideals of grade 2, and Gorenstein ideals of grade 3.

6.4 The Artin–Nagata condition and residual S_2 properties

The basic homological conditions that arise in the subject are the following.

Definition 6.6. Let $(R, \mathfrak{m})$ be a local Cohen–Macaulay ring, let $I \subseteq R$ be an ideal of grade g , and let $s \geq g$.

- (i) We say that I satisfies the *Artin–Nagata condition* AN_s if for every integer i with $g \leq i \leq s$ and every i -residual intersection J of I , the quotient R/J is Cohen–Macaulay.
- (ii) We say that I is *s -residually S_2* if for every integer i with $g \leq i \leq s$ and every i -residual intersection J of I , the ring R/J satisfies Serre’s condition S_2 .
- (iii) We say that I is *weakly s -residually S_2* if the same condition holds for every *geometric* i -residual intersection J of I .

These conditions are designed precisely to control how much of the good behavior of linkage survives in higher-codimensional residual intersections.

6.5 Determinantal examples

The theory becomes especially concrete for determinantal ideals.

Theorem 6.7 (Eisenbud–Ulrich). *Let $\mathbb{k}$ be a field of characteristic 0, let*

$$R = \mathbb{k}[x_1, \dots, x_n, y_1, \dots, y_n], \quad X = \begin{pmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \end{pmatrix},$$

and let $I = I_2(X)$ be the ideal generated by the 2×2 minors of X . Let $\mathfrak{a} = (a_1, \dots, a_\ell)$ be a minimal reduction of I , where $\ell = \ell(I)$ is the analytic spread of I . For $1 \leq i \leq \ell$, set

$$\mathfrak{a}_i = (a_1, \dots, a_i), \quad K_i = \mathfrak{a}_i : I, \quad R_i = R/K_i.$$

If $g = \text{ht}(I) = n - 1$, then for every $i \geq g + 1$ one has

$$\text{depth } R_i = 2n - i - 3 = \dim R_i - 3.$$

Moreover, I is weakly $(\ell(I) - 2)$ -residually S_2 , and the canonical module of R_i is described by a power of the image of I in R_i :

$$\omega_{R_i} \cong (IR_i)^{i-g+1}(-2n),$$

up to the usual grading convention for twists.

A similar pattern persists for maximal minors of a generic $m \times n$ matrix: the canonical module of the corresponding residual intersection is controlled by an explicit power of the image of the determinantal ideal, together with the ambient grading shift.

6.6 Generators of residual intersections

A useful structural question is to determine how a residual intersection is generated.

Proposition 6.8. *Let R be a local Cohen–Macaulay ring and let $I = (x_1, \dots, x_g) \subseteq R$ be an ideal of grade g . Let $\mathfrak{a} = (a_1, \dots, a_s) \subseteq I$, and write*

$$(a_1, \dots, a_s) = [x_1 \ \cdots \ x_g]B$$

for a $g \times s$ matrix B with entries in R . If

$$J = \mathfrak{a} : I$$

is an s -residual intersection of I , then

$$I_g(B) \subseteq J,$$

and in fact

$$J = \mathfrak{a} + I_g(B).$$

In particular, when $g = 1$ the description is immediate, and for $g > 1$ the residual intersection is obtained by adjoining the maximal minors of the coefficient matrix B .

This description is one of the standard entry points for explicit calculations.

6.7 $(g + 1)$ -residual intersections as sums of links

The first genuinely new case beyond linkage occurs for $(g + 1)$ -residual intersections.

Definition 6.9. Let I be an ideal of grade g , let $\mathfrak{a} = (a_1, \dots, a_{g+1}) \subseteq I$, and set

$$J = \mathfrak{a} : I.$$

Assume that J is a $(g + 1)$ -residual intersection of I . We say that J is a *sum of links* if

$$J = \sum_{i=1}^{g+1} (a_1, \dots, \hat{a}_i, \dots, a_{g+1}) : I.$$

Theorem 6.10 (Kustin–Ulrich). *Let R be a Gorenstein local ring and let $I \subseteq R$ be an ideal of grade g satisfying the standard residual-intersection hypotheses (for instance, in the classical situations of complete intersections, perfect ideals of grade 2, or Gorenstein ideals of grade 3). Then every $(g + 1)$ -residual intersection of I is a sum of links.*

Thus, even though a $(g + 1)$ -residual intersection need not itself be a direct link, it can still be decomposed into a controlled sum of ordinary links.

Example 6.11. Let

$$X = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix}, \quad I = I_2(X),$$

and write

$$\Delta_{ij} = x_i y_j - x_j y_i.$$

Consider the ideal

$$\mathfrak{a} = (\Delta_{12}, \Delta_{13}, \Delta_{24}, \Delta_{34}) \subseteq I.$$

Then

$$J = \mathfrak{a} : I$$

is a 4-residual intersection of the height-3 ideal I , and it decomposes as a sum of links:

$$J = (\Delta_{12}, \Delta_{13}, \Delta_{24}) : I + (\Delta_{12}, \Delta_{13}, \Delta_{34}) : I \\ + (\Delta_{12}, \Delta_{24}, \Delta_{34}) : I + (\Delta_{13}, \Delta_{24}, \Delta_{34}) : I.$$

Each summand is an ordinary link of I by one of the three-generated subideals of $\mathfrak{a}$, and their sum recovers the full residual intersection.

6.8 Closing perspective

The talk emphasized two guiding themes.

- Residual intersections provide a natural framework that extends linkage beyond the case of codimension-preserving complete intersections.
- Although the full Peskine–Szpiro package does not extend to arbitrary s -residual intersections, strong analogues survive under meaningful structural hypotheses, and determinantal ideals furnish a rich family of test cases.

In particular, the interaction among the Artin–Nagata conditions, residual S_2 properties, canonical modules, and explicit generators makes residual intersection theory a flexible bridge between homological commutative algebra and explicit geometric constructions.

7 Zach Nason — Quasi-Gorenstein Morphisms over DG-Rings

Zach Nason

Joint work with Andrew J. Soto Levins and Ryan Watson

Abstract

Quasi-Gorenstein local homomorphisms encode the Gorenstein property through finite Gorenstein dimension and a relative duality condition. The purpose of this talk is to extend this picture from ordinary local rings to commutative local DG-rings. After reviewing DG-rings, derived categories, projective dimension, and Gorenstein dimension, the discussion introduces quasi-Gorenstein morphisms of DG-rings and a Gorenstein analogue of virtual smallness. The main result characterizes quasi-Gorenstein morphisms by the behavior of DG-modules with bounded finite-length homology and finite Gorenstein dimension.

7.1 Classical quasi-Gorenstein homomorphisms

Let

$$\varphi: (R, \mathfrak{m}, k) \longrightarrow (S, \mathfrak{n}, \ell)$$

be a finite local homomorphism of Noetherian local rings. The starting point is the classical notion of a quasi-Gorenstein homomorphism.

Definition 7.1 (Quasi-Gorenstein homomorphism). The map $\varphi: R \rightarrow S$ is *quasi-Gorenstein* if

$$\mathrm{Gdim}_R S < \infty$$

and there is an isomorphism, up to suspension, in the derived category of S ,

$$\mathbf{R}\mathrm{Hom}_R(S, R) \simeq \Sigma^a S$$

for some $a \in \mathbb{Z}$.

This notion detects the Gorenstein property in two basic ways.

Proposition 7.2. *Let $(R, \mathfrak{m}, k)$ be a Noetherian local ring. The canonical map*

$$R \longrightarrow k$$

is quasi-Gorenstein if and only if R is Gorenstein.

Proposition 7.3. *If $\varphi: R \rightarrow S$ is quasi-Gorenstein, then*

$$R \text{ is Gorenstein} \iff S \text{ is Gorenstein.}$$

Moreover, for a finitely generated S -module, and more generally for a bounded homologically finite S -complex M , finiteness of Gorenstein dimension over R and over S agree:

$$\mathrm{Gdim}_R M < \infty \iff \mathrm{Gdim}_S M < \infty.$$

Thus quasi-Gorenstein maps behave as the natural morphisms along which Gorenstein finiteness is transported.

7.2 DG-rings and DG-modules

The extension of the preceding theory requires a derived replacement for a local ring.

Definition 7.4 (DG-algebra). A differential graded algebra is a graded algebra

$$R = \bigoplus_{i \in \mathbb{Z}} R_i$$

equipped with a differential

$$\partial_i: R_i \longrightarrow R_{i-1}$$

satisfying $\partial^2 = 0$ and the graded Leibniz rule

$$\partial(rs) = \partial(r)s + (-1)^{|r|}r\partial(s)$$

for homogeneous elements $r, s \in R$.

Throughout the talk the DG-algebras are strongly graded-commutative:

$$rs = (-1)^{|r||s|}sr,$$

and every homogeneous element of odd degree satisfies $r^2 = 0$.

The local DG-rings under consideration are connective and homologically finite in the following sense: $R_i = 0$ for $i < 0$, $H_0(R)$ is a Noetherian local ring, and $H(R)$ is finitely generated over $H_0(R)$. In particular, each $H_i(R)$ is a finitely generated $H_0(R)$ -module and $H_i(R) = 0$ for $i \gg 0$.

Definition 7.5 (DG-module). Let R be a DG-ring. A DG R -module is a graded

R -module

$$M = \bigoplus_{i \in \mathbb{Z}} M_i$$

with a differential $\partial_M: M_i \rightarrow M_{i-1}$ satisfying

$$\partial_M(rm) = \partial_R(r)m + (-1)^{|r|}r\partial_M(m)$$

for homogeneous $r \in R$ and $m \in M$.

A morphism of local DG-rings

$$\varphi: R \longrightarrow S$$

is required to induce a local homomorphism

$$H_0(R) \longrightarrow H_0(S).$$

7.3 The derived category and derived functors

Write $\mathbf{D}(R)$ for the derived category of DG R -modules. Quasi-isomorphic DG-modules become isomorphic in $\mathbf{D}(R)$. The two basic derived functors are

$$- \otimes_R^{\mathbf{L}} - \quad \text{and} \quad \mathbf{R}\mathrm{Hom}_R(-, -).$$

They may be computed by replacing the relevant argument with a suitable semifree, projective, or injective resolution. Thus, if

$$P \xrightarrow{\sim} M, \quad F \xrightarrow{\sim} M, \quad N \xrightarrow{\sim} I$$

are appropriate resolutions, then the ordinary tensor and Hom constructions on these replacements compute the corresponding derived functors.

The cohomology of derived Hom recovers Ext:

$$H^i(\mathbf{R}\mathrm{Hom}_R(M, N)) \cong \mathrm{Ext}_R^i(M, N),$$

with the usual conversion between homological and cohomological grading conventions.

7.4 Projective dimension and Gorenstein dimension

One way to formulate projective dimension for a DG-module M is through eventual Ext-vanishing. In the bounded setting,

$$\mathrm{projdim}_R(M) = \inf \left\{ n \in \mathbb{Z} \mid \mathrm{Ext}_R^i(M, N) = 0 \text{ for all } N \in \mathbf{D}_b(R) \text{ and all } i > n - \inf(N) \right\}.$$

Equivalently, finite projective dimension means that the functor

$$\mathbf{R}\mathrm{Hom}_R(M, -)$$

has finite homological displacement.

Gorenstein dimension is defined using derived biduality.

Definition 7.6 (Derived reflexivity). Let $M \in \mathbf{D}_b^f(R)$. The DG-module M is *derived reflexive* if

$$\mathbf{R}\mathrm{Hom}_R(M, R) \in \mathbf{D}_b^f(R)$$

and the canonical biduality morphism

$$M \longrightarrow \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(M, R), R)$$

is an isomorphism in $\mathbf{D}(R)$.

Definition 7.7 (Gorenstein dimension). For a derived-reflexive DG-module $M \in \mathbf{D}_b^f(R)$, set

$$\mathrm{Gdim}_R(M) = -\inf \mathbf{R}\mathrm{Hom}_R(M, R).$$

If M is not derived reflexive, then

$$\mathrm{Gdim}_R(M) = \infty.$$

This is the DG analogue of the Auslander–Bridger Gorenstein dimension for finitely generated modules over a Noetherian local ring.

7.5 Quasi-Gorenstein morphisms of DG-rings

The classical definition extends directly once Gorenstein dimension and derived Hom have been set up in the DG category.

Definition 7.8 (Quasi-Gorenstein morphism). Let

$$\varphi: R \longrightarrow S$$

be a morphism of commutative local DG-rings. Assume

$$\mathrm{Gdim}_R S < \infty.$$

Then φ is *quasi-Gorenstein* if

$$\mathbf{R}\mathrm{Hom}_R(S, R) \simeq \Sigma^a S$$

in $\mathbf{D}(S)$ for some $a \in \mathbb{Z}$.

Thus the relative dualizing object $\mathbf{R}\mathrm{Hom}_R(S, R)$ is required to be nothing more than a shift of the target DG-ring itself. When R and S are ordinary Noetherian local rings and φ is module-finite, this recovers the classical definition.

7.6 Thick subcategories and virtual smallness

The main characterization is phrased in terms of thick subcategories of the derived category.

Definition 7.9 (Thick subcategory). A full subcategory $\mathcal{C} \subseteq \mathbf{D}(R)$ is *thick* if it is closed under isomorphisms, shifts, finite direct sums, direct summands, and distinguished triangles.

For $M \in \mathbf{D}(R)$, write

$$\mathrm{Thick}_R(M)$$

for the smallest thick subcategory of $\mathbf{D}(R)$ containing M .

Two basic examples are important:

$$\mathrm{Thick}_R(R) = \{\text{perfect DG } R\text{-modules}\},$$

and, in the local Noetherian setting,

$$\mathrm{Thick}_R(k) = \{M \in \mathbf{D}_b^f(R) : H(M) \text{ has finite length}\}.$$

Definition 7.10 (Virtually small). An object $M \in \mathbf{D}(R)$ is *virtually small* if

$$\mathrm{Thick}_R(M) \cap \mathrm{Thick}_R(R) \neq \{0\}.$$

Equivalently, M finitely builds a nonzero perfect object.

For the Gorenstein analogue, replace perfect objects by objects of finite Gorenstein dimension.

Definition 7.11 (Virtually G -small). Let $M \in \mathbf{D}_b^f(R)$. The DG-module M is *virtually*

G -small if there exists a nonzero object

$$F_M \in \text{Thick}_R(M)$$

with

$$\text{Gdim}_R(F_M) < \infty.$$

Equivalently, the thick subcategory generated by M meets the class of nonzero derived-reflexive objects.

This definition is designed so that finite Gorenstein dimension plays the role that finite projective dimension plays in ordinary virtual smallness.

7.7 The complete-intersection analogy

The motivation for the main theorem comes from the corresponding detection theorem for complete-intersection maps. A result of Briggs, Iyengar, Letz, and Pollitz gives a model for what one should expect.

Theorem 7.12 (Briggs–Iyengar–Letz–Pollitz). *Let $\varphi: R \twoheadrightarrow S$ be a surjective local homomorphism of Noetherian local rings of finite flat dimension. Then φ is a complete-intersection homomorphism if and only if every S -complex with bounded finite-length homology that is proxy small over R is proxy small over S .*

The guiding analogy is therefore

finite projective dimension	$\longleftrightarrow$	finite Gorenstein dimension,
complete-intersection map	$\longleftrightarrow$	quasi-Gorenstein map,
proxy/virtual smallness	$\longleftrightarrow$	virtual G -smallness.

7.8 The main characterization

Let

$$\mathbf{D}_b^{\text{fl}}(S) = \left\{ M \in \mathbf{D}_b^f(S) \mid H(M) \text{ has finite length over } H_0(S) \right\}.$$

The central theorem identifies quasi-Gorenstein morphisms exactly through the virtual G -smallness of such finite-length objects.

Theorem 7.13 (Nason–Soto Levins–Watson, 2026). *Let*

$$\varphi: R \longrightarrow S$$

be a morphism of commutative local DG-rings such that S_0 is a Noetherian local ring and

$$\mathrm{Gdim}_R S < \infty.$$

Then the following are equivalent:

- (i) φ is quasi-Gorenstein;
- (ii) every DG S -module

$$M \in \mathbf{D}_b^{\mathrm{fl}}(S)$$

with

$$\mathrm{Gdim}_R M < \infty$$

is virtually G -small over S .

The forward implication reflects the preservation of finite Gorenstein dimension along a quasi-Gorenstein morphism: an S -object having finite Gorenstein dimension over R also has finite Gorenstein dimension over S , so it already supplies the required finite-Gorenstein-dimension object in its own thick closure. The converse is the substantive direction: the finite-length test objects force the relative dualizing complex

$$\mathbf{R}\mathrm{Hom}_R(S, R)$$

to have the rigidity required to be a shift of S .

The theorem is new even when both R and S are ordinary rings. It gives a Gorenstein counterpart to the complete-intersection detection theorem above and shows that virtual G -smallness is the correct categorical finiteness condition for quasi-Gorenstein morphisms.

Summary of the Lecture

The central object is a morphism of local DG-rings

$$\varphi: R \longrightarrow S$$

with finite relative Gorenstein dimension. Such a morphism is quasi-Gorenstein precisely when the relative dualizing DG-module

$$\mathbf{R}\mathrm{Hom}_R(S, R)$$

is a shift of S . The classical consequences—detection of Gorenstein rings and transfer of finite Gorenstein dimension—persist in the DG setting. The main theorem then characterizes the quasi-Gorenstein property through finite-length DG S -modules: finite Gorenstein dimension over R must force virtual G -smallness over S . This is the Gorenstein counterpart of proxy-small detection for complete intersections.

8 Maria Akter — Characterizations of Minimal Frobenius Splitting Numbers in Gorenstein Rings

Maria Akter

Frobenius splitting numbers, splitting ideals, and hypersurface perturbations

Abstract

For an F -finite local ring of characteristic $p > 0$, the Frobenius splitting number $a_e(R)$ is the maximal number of free R -summands occurring in $F_*^e R$. The lecture studies the smallest possible positive values of these numbers in the F -pure Gorenstein setting. The central result characterizes minimal Frobenius splitting numbers in terms of the Frobenius splitting ideals $I_e(R)$ and the splitting prime $\mathcal{P}(R)$. In a Gorenstein ring, the splitting ideals admit a concrete description by colon ideals associated to a system of parameters and a socle generator. This description leads to an explicit hypersurface criterion and to a perturbation result for quotients $R/(f + \epsilon)$. A Veronese example shows that the Gorenstein hypothesis is essential for the propagation of minimal splitting numbers across Frobenius iterates.

8.1 Frobenius and restriction of scalars

Throughout this lecture, $(R, \mathfrak{m}, k)$ is a commutative Noetherian local ring of characteristic $p > 0$, and

$$d = \dim R.$$

For each $e \geq 1$, write $q = p^e$. The e th iterate of the Frobenius endomorphism is

$$F^e: R \longrightarrow R, \quad r \longmapsto r^q.$$

The module $F_*^e R$ is obtained from R by restriction of scalars along F^e . Thus $F_*^e R$ agrees with R as an abelian group, but its R -module structure is given by

$$r \cdot F_*^e s = F_*^e(r^q s) \quad (r, s \in R).$$

Definition 8.1 (F -finite ring). The ring R is F -finite if $F_* R$ is a finitely generated R -module. Equivalently, $F_*^e R$ is finitely generated for every $e \geq 1$.

Example 8.2 (Polynomial rings over a perfect field). Let

$$R = k[x_1, \dots, x_d]$$

with k perfect. Then $F_*^e R$ is free of rank $q^d = p^{ed}$, with R -basis

$$\left\{ F_*^e(x_1^{i_1} \cdots x_d^{i_d}) \mid 0 \leq i_j < q \right\}.$$

Thus, in the regular case, every summand of $F_*^e R$ is free.

The preceding example is a special case of Kunz's theorem.

Theorem 8.3 (Kunz). *Let R be a Noetherian local ring of characteristic $p > 0$. Then R is regular if and only if Frobenius is flat. In the F -finite local setting, this is equivalent to $F_*^e R$ being a free R -module for some, equivalently every, $e \geq 1$.*

8.2 F -purity, strong F -regularity, and splitting numbers

The next two singularity classes are defined by the existence of splittings of Frobenius.

Definition 8.4 (F -purity). An F -finite local ring R is F -pure if the Frobenius map

$$R \longrightarrow F_* R, \quad 1 \longmapsto F_* 1,$$

splits as a map of R -modules. Equivalently, there is an R -linear map

$$\phi: F_* R \longrightarrow R$$

with $\phi(F_* 1) = 1$.

Definition 8.5 (Strong F -regularity). Put

$$R^\circ = R \setminus \bigcup_{\mathfrak{p} \in \text{Min } R} \mathfrak{p}.$$

The ring R is *strongly F -regular* if, for every $c \in R^\circ$, there is an integer $e \geq 1$ such that the map

$$R \longrightarrow F_*^e R, \quad 1 \longmapsto F_*^e c,$$

splits. Equivalently, there is some

$$\phi \in \text{Hom}_R(F_*^e R, R)$$

for which $\phi(F_*^e c) = 1$.

Even when R is singular, $F_*^e R$ may contain a substantial free direct summand. This leads to the numerical invariants at the center of the lecture.

Definition 8.6 (Frobenius splitting numbers). For every $e \geq 1$, write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e,$$

where M_e has no free direct summand and $a_e(R)$ is maximal. The integer $a_e(R)$ is the e th *Frobenius splitting number* of R .

When the residue field is perfect, the F -signature is

$$s(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{ed}}.$$

The limit exists for F -finite local rings. The fundamental extremal properties are

$$s(R) > 0 \iff R \text{ is strongly } F\text{-regular,}$$

and

$$s(R) = 1 \iff R \text{ is regular.}$$

Thus the free rank of Frobenius detects both regularity and strong F -regularity asymptotically.

There is also an asymptotic invariant adapted to F -pure rings that are not strongly F -regular. If $\mathcal{P}(R)$ denotes the splitting prime defined below and k is perfect, set

$$n = \dim(R/\mathcal{P}(R)).$$

This integer is the *Frobenius splitting dimension*. The corresponding splitting ratio is

$$r_F(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{en}},$$

and for an F -pure ring one has

$$0 < r_F(R) \leq 1.$$

When $n = d$, the splitting prime is zero and the splitting ratio agrees with the F -signature.

8.3 Splitting ideals and the splitting prime

The splitting numbers can be recovered from a family of ideals measuring which elements fail to produce a Frobenius splitting.

Definition 8.7 (Frobenius splitting ideal). For $e \geq 1$, define

$$I_e(R) = \{r \in R \mid \phi(F_*^e r) \in \mathfrak{m} \text{ for every } \phi \in \operatorname{Hom}_R(F_*^e R, R)\}.$$

Thus $r \notin I_e(R)$ precisely when some R -linear map

$$\phi: F_*^e R \longrightarrow R$$

sends $F_*^e r$ to a unit. After rescaling ϕ , this is equivalent to the map

$$R \longrightarrow F_*^e R, \quad 1 \longmapsto F_*^e r,$$

splitting.

A standard length formula relates $I_e(R)$ to the free rank:

$$a_e(R) = [k^{1/q} : k] \ell_R(R/I_e(R)), \quad q = p^e. \quad (8.1)$$

In particular, if R is F -pure, then the smallest possible value of $a_e(R)$ is

$$[k^{1/q} : k],$$

and this minimum occurs exactly when

$$I_e(R) = \mathfrak{m}.$$

When k is perfect, the minimum is simply 1.

Definition 8.8 (Splitting prime). Assume that R is F -pure. The *splitting prime* is

$$\mathcal{P}(R) = \bigcap_{e \geq 1} I_e(R).$$

It is a prime ideal. Moreover,

$$R \text{ is strongly } F\text{-regular} \iff \mathcal{P}(R) = 0.$$

At the opposite extreme, the condition

$$\mathcal{P}(R) = \mathfrak{m}$$

means that the Frobenius splitting dimension is zero. The main theorem identifies this condition with minimal Frobenius splitting numbers in the Gorenstein setting.

8.4 The Gorenstein colon formula

Suppose now that $(R, \mathfrak{m}, k)$ is a d -dimensional Gorenstein local ring. Let

$$\underline{x} = x_1, \dots, x_d$$

be a system of parameters and write

$$(\underline{x}) = (x_1, \dots, x_d).$$

Since R is Cohen–Macaulay, $R/(\underline{x})$ is Artinian Gorenstein, so its socle is one-dimensional over k . Choose $u \in R$ whose image generates

$$\text{soc}(R/(\underline{x})).$$

Equivalently,

$$(\underline{x}) :_R \mathfrak{m} = (\underline{x}, u). \quad (8.2)$$

For an ideal $J \subseteq R$ and $q = p^e$, write

$$J^{[q]} = (r^q : r \in J).$$

The Gorenstein hypothesis gives an explicit description of every splitting ideal.

Proposition 8.9 (Splitting ideals in the Gorenstein case). *With the notation above, for every $e \geq 1$ and $q = p^e$,*

$$I_e(R) = ((\underline{x})^{[q]} :_R u^q). \quad (8.3)$$

This formula is the computational input behind the hypersurface statements later in the lecture. It translates a condition involving all Cartier maps

$$F_*^e R \longrightarrow R$$

into a single colon ideal.

8.5 Hypersurface quotients

The familiar Fedder criterion is the basic model for the hypersurface calculations.

Criterion 8.10 (Fedder, hypersurface form). *Let $(S, \mathfrak{n}, k)$ be an F -finite regular local ring of characteristic $p > 0$, and let $0 \neq f \in \mathfrak{n}$. Then*

$$S/(f) \text{ is } F\text{-pure} \iff f^{p-1} \notin \mathfrak{n}^{[p]}.$$

For a Gorenstein ambient ring, the corresponding statement is encoded by the splitting ideals. Let $f \in R$ be a nonzerodivisor. A Frobenius map on $R/(f)$ is obtained from a map on R by multiplication by

$$f^{q-1}.$$

More precisely, the map determining splitting on the quotient is represented upstairs by

$$F_*^e R \xrightarrow{F_*^e(f^{q-1}, -)} F_*^e R \xrightarrow{\phi} R.$$

Consequently,

$$I_e(R/(f)) = \frac{(I_e(R) :_R f^{q-1})}{(f)}. \quad (8.4)$$

Combining (8.4) with the Gorenstein colon formula (8.3) gives

$$I_e(R/(f)) = \frac{(((\underline{x})^{[q]} :_R u^q) :_R f^{q-1})}{(f)}. \quad (8.5)$$

This is the precise form in which the Gorenstein structure enters the quotient problem.

8.6 Minimal Frobenius splitting numbers

The central result of the lecture gives several equivalent formulations of the smallest possible positive Frobenius splitting numbers.

Theorem 8.11 (Akter: characterization of minimal splitting numbers). *Let $(R, \mathfrak{m}, k)$ be an F -finite, F -pure, Gorenstein local ring of characteristic $p > 0$. Then the following are equivalent.*

(1)

$$a_1(R) = [k^{1/p} : k].$$

(2) *There exists $e \geq 1$ such that*

$$a_e(R) = [k^{1/p^e} : k].$$

(3) *For every $e \geq 1$,*

$$a_e(R) = [k^{1/p^e} : k].$$

(4)

$$I_1(R) = \mathfrak{m}.$$

(5) *There exists $e \geq 1$ such that*

$$I_e(R) = \mathfrak{m}.$$

(6) *For every $e \geq 1$,*

$$I_e(R) = \mathfrak{m}.$$

(7)

$$\mathcal{P}(R) = \mathfrak{m}.$$

If k is perfect, conditions (1)–(3) simply say, respectively,

$$a_1(R) = 1, \quad a_e(R) = 1 \text{ for some } e, \quad a_e(R) = 1 \text{ for every } e.$$

Structure of the proof. The equivalence between the splitting-number conditions and the corresponding splitting-ideal conditions follows immediately from (8.1): the positive minimum of $a_e(R)$ is attained exactly when

$$\ell_R(R/I_e(R)) = 1,$$

which is equivalent to $I_e(R) = \mathfrak{m}$.

The genuinely Gorenstein step is the propagation of the condition $I_e(R) = \mathfrak{m}$ from one Frobenius level to all levels. In the Gorenstein case, the Cartier module

$$\mathrm{Hom}_R(F_*^e R, R)$$

is cyclic over $F_*^e R$, and its generators may be chosen compatibly under Frobenius iteration. This compatibility forces maximality of one splitting ideal to propagate to $I_1(R)$ and then to every $I_e(R)$. Finally,

$$\mathcal{P}(R) = \bigcap_{e \geq 1} I_e(R),$$

so condition (6) implies (7), while (7) forces every $I_e(R)$ to equal $\mathfrak{m}$ because each $I_e(R)$ is contained in $\mathfrak{m}$. $\square$

There is a useful elementary inequality among the splitting numbers that explains one direction of the theorem without using Gorenstein duality.

Lemma 8.12 (Supermultiplicativity of splitting numbers). *For all $e, f \geq 1$,*

$$a_{e+f}(R) \geq a_e(R)a_f(R).$$

Proof. Write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e.$$

Applying F_*^f gives

$$F_*^{e+f} R \cong (F_*^f R)^{\oplus a_e(R)} \oplus F_*^f M_e.$$

Since each copy of $F_*^f R$ contains $R^{\oplus a_f(R)}$ as a direct summand, the first term contributes at least

$$a_e(R)a_f(R)$$

free summands to $F_*^{e+f}R$. □

Corollary 8.13. *Assume that R is F -pure and that k is perfect. If*

$$a_e(R) = 1$$

for some $e \geq 1$, then

$$a_1(R) = 1.$$

Proof. By Lemma 8.12,

$$a_e(R) \geq a_1(R)^e.$$

Since F -purity gives $a_1(R) \geq 1$, the equality $a_e(R) = 1$ forces $a_1(R) = 1$. □

The converse implication

$$a_1(R) = 1 \implies a_e(R) = 1 \text{ for all } e$$

is therefore the subtle direction, and it is precisely where the Gorenstein hypothesis is used.

8.7 A hypersurface criterion

The main theorem combines with the colon formula (8.5) to give an explicit criterion for hypersurface quotients.

Corollary 8.14 (Minimal splitting numbers for a hypersurface). *Let $(R, \mathfrak{m}, k)$ be an F -finite, F -pure, Gorenstein local ring of dimension d and characteristic $p > 0$. Let $f \in \mathfrak{m}$ be a nonzerodivisor. Choose a system of parameters*

$$\underline{x} = x_1, \dots, x_d$$

and an element $u \in R$ satisfying

$$(\underline{x}) :_R \mathfrak{m} = (\underline{x}, u).$$

Then the following are equivalent.

(1)

$$a_1(R/(f)) = [k^{1/p} : k].$$

(2) *There exists $e \geq 1$ such that*

$$a_e(R/(f)) = [k^{1/p^e} : k].$$

(3) For every $e \geq 1$,

$$a_e(R/(f)) = [k^{1/p^e} : k].$$

(4)

$$\left(((\underline{x})^{[p]} :_R u^p) :_R f^{p-1} \right) = \mathfrak{m}.$$

(5) There exists $e \geq 1$ such that

$$\left(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1} \right) = \mathfrak{m}.$$

(6) For every $e \geq 1$,

$$\left(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1} \right) = \mathfrak{m}.$$

(7) The splitting prime of $R/(f)$ is the maximal ideal:

$$\mathcal{P}(R/(f)) = \mathfrak{m}/(f).$$

Moreover, if these equivalent conditions hold and

$$\epsilon \in (\underline{x})^{[p]},$$

then

$$\mathcal{P}(R/(f + \epsilon)) = \mathfrak{m}/(f + \epsilon).$$

The final assertion gives a particularly concrete persistence statement. Once the quotient has zero splitting dimension, perturbing f by an element in the Frobenius power of the parameter ideal preserves that property.

8.8 Perturbations and continuity

The hypersurface result fits into a broader perturbation problem: how much of a singularity is unchanged after replacing a defining equation f by $f + \epsilon$ with ϵ very deep in the maximal ideal?

A classical theorem of Samuel provides a strong answer for isolated hypersurface singularities. Let

$$S = k[[x_1, \dots, x_d]], \quad \mathfrak{n} = (x_1, \dots, x_d),$$

and let

$$J(f) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right)$$

be the Jacobian ideal. If $J(f)$ is $\mathfrak{n}$ -primary and

$$\epsilon \in \mathfrak{n}J(f)^2,$$

then there is an automorphism of S carrying f to $f + \epsilon$. Consequently,

$$S/(f) \cong S/(f + \epsilon).$$

Exact isomorphism is much stronger than what is generally available, so one may instead ask whether numerical invariants vary continuously. The F -signature has this property at reduced parameter elements: in the Cohen–Macaulay F -finite setting, Polstra and Smirnov proved that for every $\delta > 0$, sufficiently small $\mathfrak{m}$ -adic perturbations satisfy

$$\left| s(R/(f)) - s(R/(f + \epsilon)) \right| < \delta.$$

The results above sharpen this perspective in the minimal-splitting situation: under the Gorenstein hypotheses, one obtains exact persistence of the splitting-prime condition for the explicit class of perturbations

$$\epsilon \in (\underline{x})^{[p]}.$$

8.9 Why the Gorenstein hypothesis matters

The propagation in Theorem 8.11 fails without the Gorenstein assumption. The lecture illustrates this with a Veronese ring.

Example 8.15 (A non-Gorenstein Veronese ring). Let k be a perfect field of characteristic 2, let

$$S = k[x, y],$$

and let

$$R = S^{(3)} = \bigoplus_{n \geq 0} S_{3n}$$

be the third Veronese subring, localized at its homogeneous maximal ideal when viewed in the local setting. As an R -module, the polynomial ring decomposes according to total degree modulo 3:

$$S = R \oplus M_1 \oplus M_2,$$

where

$$\begin{aligned} M_1 &= \bigoplus_{a+b \equiv 1 \pmod{3}} kx^a y^b, \\ M_2 &= \bigoplus_{a+b \equiv 2 \pmod{3}} kx^a y^b. \end{aligned}$$

Thus R is a direct summand of the regular ring S , and in this tame situation it is strongly F -regular.

The Goto–Watanabe criterion for Veronese subrings says that the n th Veronese of a polynomial ring in d variables is Gorenstein precisely when n divides d . Here

$$n = 3, \quad d = 2,$$

so R is not Gorenstein.

The splitting-number computation recorded in the lecture may be expressed as

$$a_e(R) = \# \left\{ (a, b) \in \{0, \dots, 2^e - 1\}^2 \mid a + b \equiv 0 \pmod{3} \right\}.$$

For $e = 1$, the only admissible pair is $(0, 0)$, so

$$a_1(R) = 1.$$

For $e = 2$, the admissible pairs have sums 0, 3, or 6. There are respectively 1, 4, and 1 such pairs, giving

$$a_2(R) = 6.$$

Hence

$$a_1(R) = 1 \quad \text{but} \quad a_2(R) > 1.$$

Thus the implication from a minimal first splitting number to minimal splitting numbers at every Frobenius level is false without the Gorenstein hypothesis.

Remark 8.16. The inequality in Lemma 8.12 is perfectly compatible with this example: it only gives

$$a_2(R) \geq a_1(R)^2 = 1.$$

The Gorenstein structure supplies the additional rigidity needed to force equality at all levels.

Summary of the Lecture

The lecture organizes the minimal Frobenius splitting problem around three equivalent types of data.

(i) The decomposition

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e$$

produces the Frobenius splitting numbers. Their smallest positive value is

$$[k^{1/p^e} : k],$$

which equals 1 when the residue field is perfect.

(ii) The splitting ideal

$$I_e(R) = \{r \in R : \phi(F_*^e r) \in \mathfrak{m} \text{ for every } \phi \in \text{Hom}_R(F_*^e R, R)\}$$

converts the numerical condition into ideal theory through

$$a_e(R) = [k^{1/p^e} : k] \ell_R(R/I_e(R)).$$

Thus $a_e(R)$ is minimal exactly when $I_e(R) = \mathfrak{m}$.

(iii) In a Gorenstein ring, if u is a socle generator modulo a parameter ideal $(\underline{x})$, then

$$I_e(R) = ((\underline{x})^{[p^e]} :_R u^{p^e}).$$

This cyclic Gorenstein description forces minimality at one Frobenius level to propagate to all levels and identifies it with

$$\mathcal{P}(R) = \mathfrak{m}.$$

For a hypersurface quotient, multiplication by f^{p^e-1} yields the explicit criterion

$$I_e(R/(f)) = \frac{(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1})}{(f)}.$$

This leads to the perturbation statement for $f + \epsilon$. The third Veronese example in characteristic 2 shows that the Gorenstein hypothesis is not cosmetic: one can have $a_1(R) = 1$ but $a_2(R) = 6$ in a strongly F -regular non-Gorenstein ring.